



# EUROPEAN **MiCROSOFT** **FABRiC** Community Conference

VIENNA 15-18 SEPTEMBER 2025

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#FABCONEUROPE25





# We're at capacity, now what?

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# ABOUT ME



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*Over 10 years of (Microsoft) Data & AI experience  
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# Capacity Basics



# Fabric capacities & capacity based pricing



One capacity to power all your Fabric computes



Easier-to-read invoice



One capacity = one price



More predictable monthly costs



Less need for scaling of components

# fabric capacity = highway

1 lane = 1 CU \*

Current highway = F4



\* Not really, for illustration purposes only

# Two operation types



## Background operations

*Longer running, typically scheduled, operations such as semantic model or dataflow refreshes, spark jobs or data pipelines*

*E.g. Data Factory Pipeline, PySpark notebook, ...*



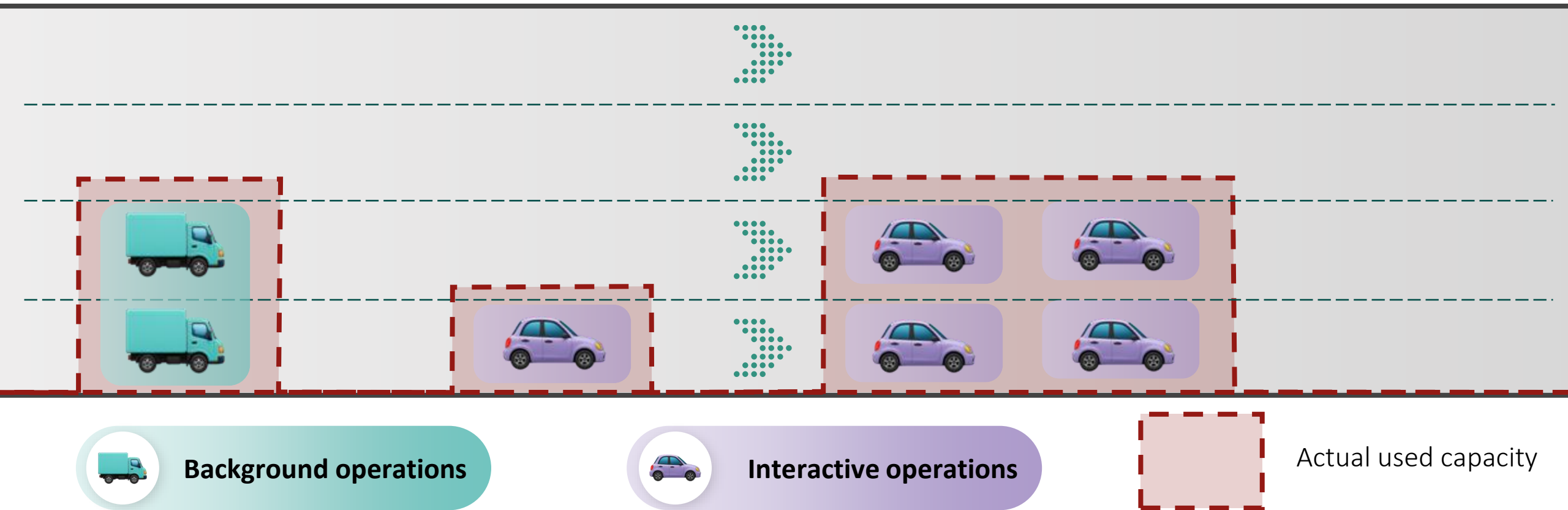
## Interactive operations

*On-demand requests and operations that can be triggered by user interactions with the UI, such as data model queries generated by report visuals*

*E.g. Power BI render, DirectQuery, ...*

*Background operations get priority over interactive operations*

# fabric capacity = highway





Bursting



Smoothing

- Concepts introduced by Microsoft to handle *flexibility* within the fixed capacity
- Allows you to go over capacity when needed and pay back later



## Bursting

Enables you to (temporarily) use more capacity than you have purchased. This to cover bursts (e.g. nightly load, heavy query, ...) without the need to scale.  
= temporarily opening some extra lanes on your highway



Background operations

Actual moment when operation happens



## Smoothing

Smoothing flattens the compute usage over time (24hrs) to help you avoid being penalized for short bursts.  
= less room for other traffic in the next 24hrs



Background operations

*Usage smoothed over 24hrs*



## Background operations

- Power BI
  - Scheduled Refresh
  - Report subscription
  - Interactive queries over Direct Lake
- Data Factory (& dataflows Gen2)
- Warehouse
- Lakehouse
- Notebook
- Spark Job
- OneLake
- Real-time analytics (EventStream)
- CoPilot

Smoothed over 24 h



## Interactive operations

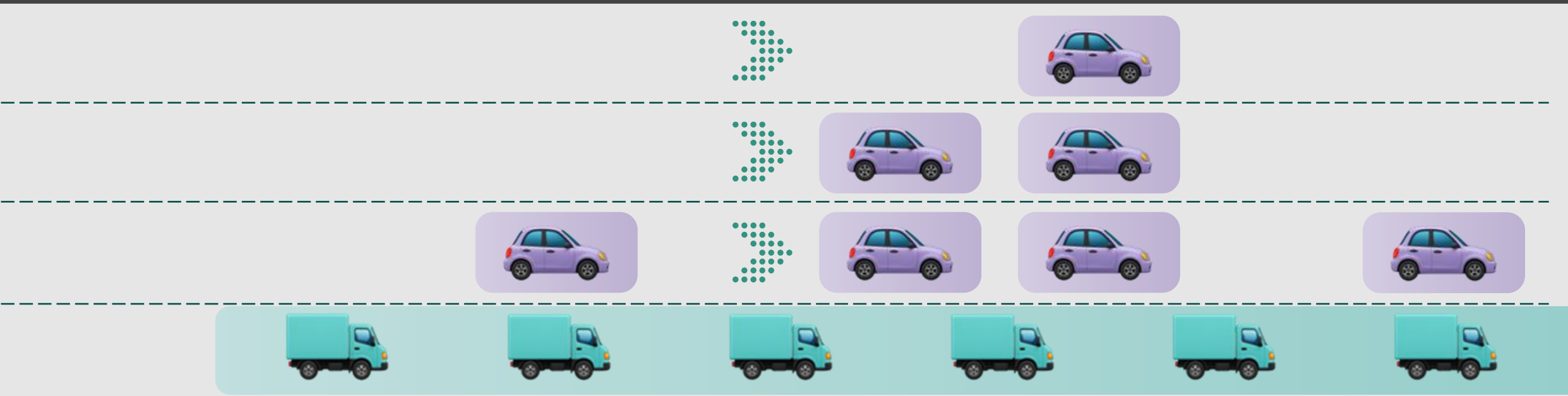
- Power BI
  - Render
  - DirectQuery
  - Interactive queries to semantic model
  - Web modelling
- KQL uptime
- Fabric API for GraphQL
- Fabric Database SQL queries
- AI Functions in Power BI
- User Data Functions (UDF)

Smoothed over 5 mins

*Background operations get priority over interactive operations*

# Background & interactive operations

All Fabric operations fill your capacity





**Background operations**

Smoothed 24hrs



**Interactive operations**

smoothed 5 mins



Bursting



Smoothing

- Concepts introduced by Microsoft to handle *flexibility* within the fixed capacity
  - Allows you to go over capacity when needed and pay back later
- you get further with lower capacities
- No need for (constant) scaling

# We're at capacity



# We're at capacity



1. When/Why does it occur
2. How do we notice
3. Identifying what is the cause

# When/why does it occur?

- Build up of (background) activities
  - Heavy (interactive) queries
  - Long-running semantic models
  - Longer-than-usual running ETL jobs
  - Rerunning jobs or refreshes
  - Amount of operations not fit for capacity
  - ...



## Smoothed background operations build up

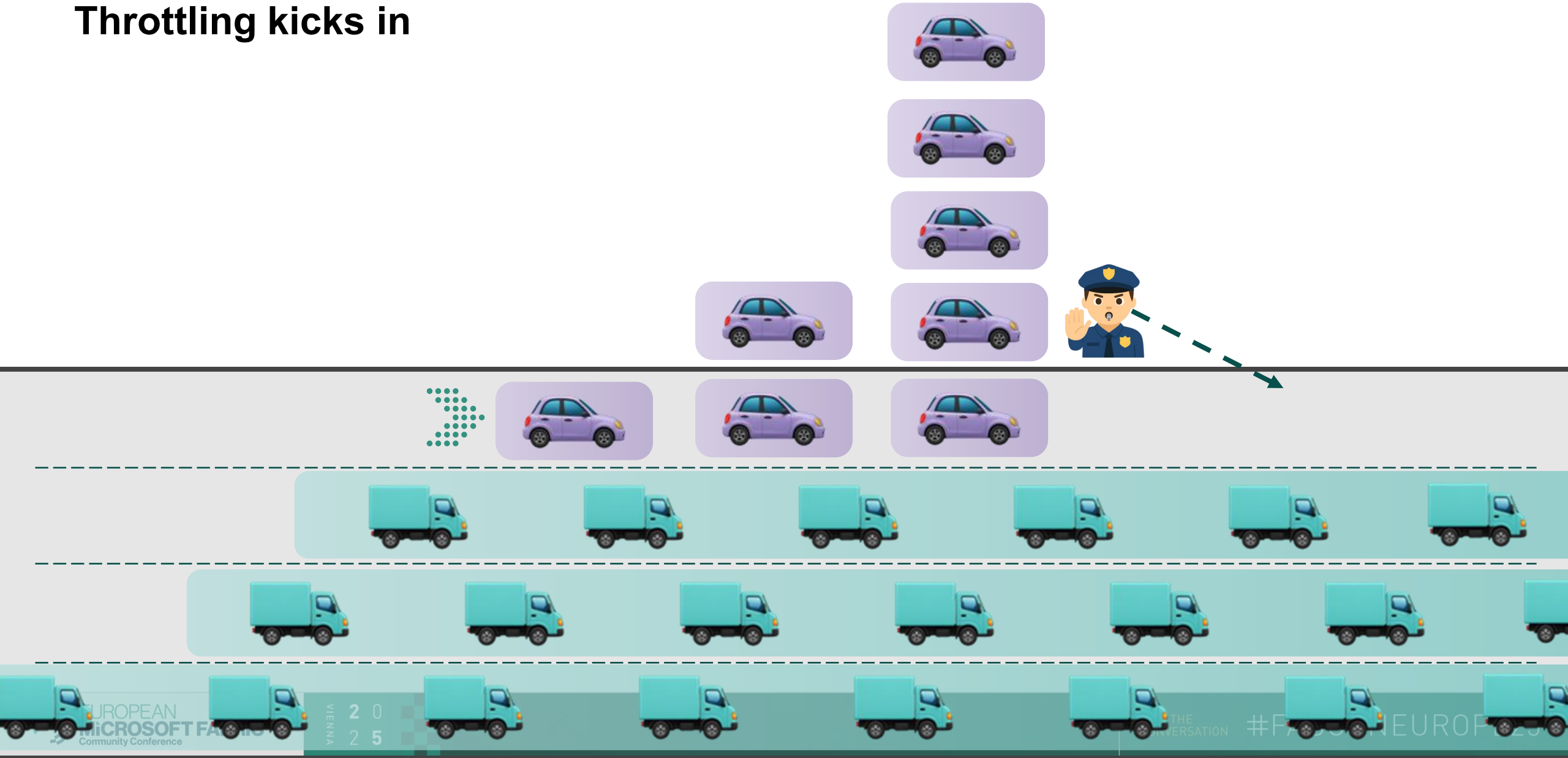


## Interactive operations

smoothed 5 mins

# Background & interactive operations

Throttling kicks in



# When does it occur?

**A capacity limit is hit when there is no more future capacity to “pay back” the over usage (*overage*)**

Sustained over usage results in throttling = penalty for over-use

1. Interactive requests first are **delayed**

- When smoothed usage exceeds 10 minutes of 100% Capacity



2. Interactive requests are **rejected** 

- When smoothed usage exceeds 60 minutes of 100% Capacity



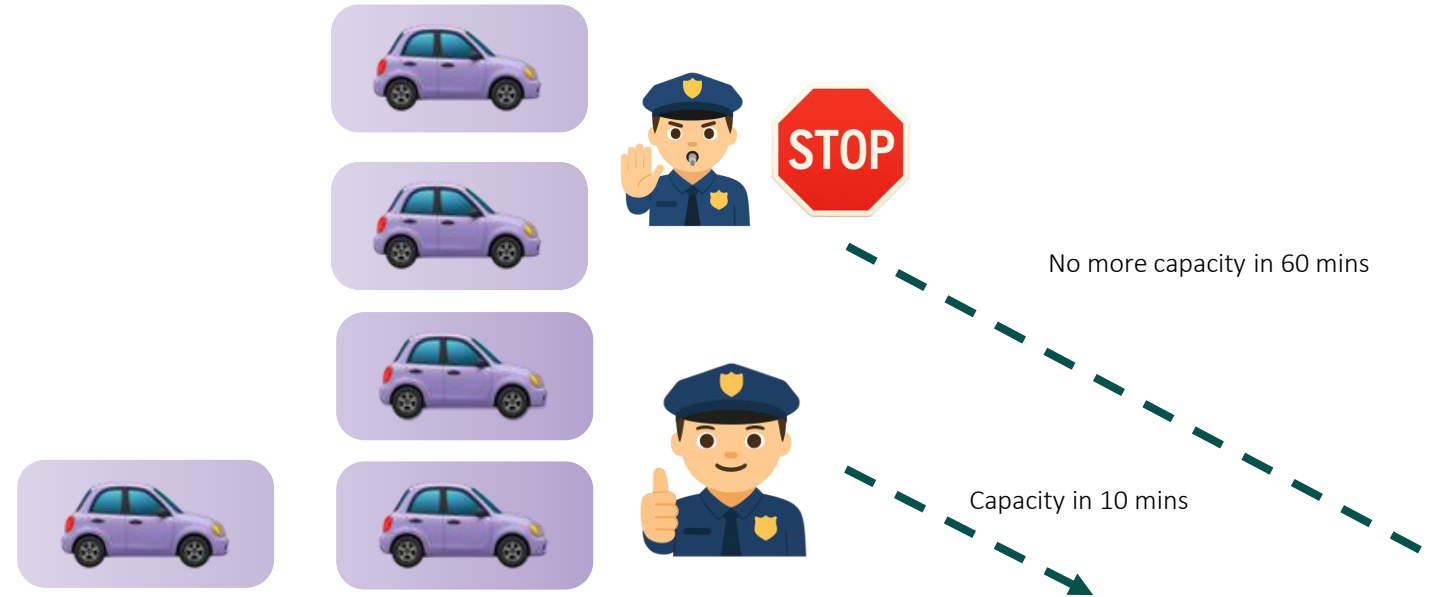
3. Background requests are **rejected** 

- When smoothed usage exceeds 24 hours of 100% Capacity



# Background & interactive operations

Throttling kicks in

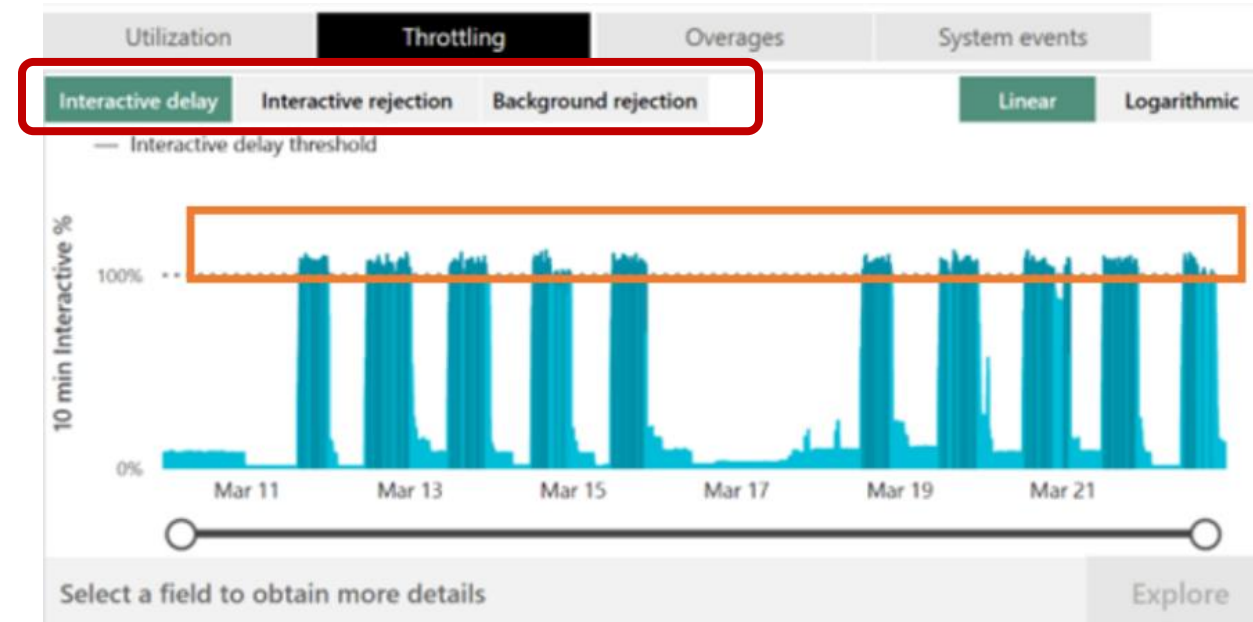


# When does it occur?

A capacity limit is hit when there is no more future capacity to “pay back” the over usage (*overage*)

Over usage can be monitored in capacity metrics app

1. Interactive requests first are **delayed**
  - When smoothed usage exceeds 10 minutes of 100% Capacity
2. Interactive requests are **rejected** 
  - When smoothed usage exceeds 60 minutes of 100% Capacity
3. Background requests are **rejected** 
  - When smoothed usage exceeds 24 hours of 100% Capacity



# How do we notice?



**New Eventhouse**

⚠️

CapacityLimitExceeded.; SessionId -----

**Unable to connect**

We encountered an error while trying to connect.

Details: "Unable to complete the action because your organization's Fabric compute capacity has exceeded its limits. Try again later."

**Failed to load**

Unable to complete the action because your organization's Fabric...

**Failed to load**

Unable to complete the action because your organization's Fabric compute capacity has exceeded its limits. Try again later.

**We couldn't create a notebook**

Unable to complete the action because your organization's Fabric compute capacity has exceeded its limits. Try again later.

Show details ▾

Cluster URI <https://wabi-europe-north-b-redirect.analysis.windows.net/>

[Help](#)

Copy to clipboard

Get help Close

**Unable to load model due to reaching capacity limits**

Unable to open this report because your organization's compute capacity has exceeded its limits. Try again later.

Close

Source: Fabric Community

# How to identify what is going wrong



Check what is running & failing  
= assess impact



Monitor tab



Check the cause



Capacity  
Metrics App



Detailed assessment of other  
risk factors to avoid future  
overruns



FUAM

# How to identify what is going wrong?

## 1. Fabric Monitor tab



**Power BI Monitor**

Search

Home Create Browse OneLake catalog Apps Metrics **Monitor** Learn Real-Time Workloads Workspaces My workspace

### Monitor

View and track the status of the activities across all the workspaces for which you have permissions within Microsoft Fabric.

[Refresh](#)

[Filter by keyword](#) [Filter](#) [Column Options](#)

[Clear all](#) To apply filters, select the values from the Filter dropdown menu.

| Activity name                        | Status      | Item type      | Start time           | Submitted by | Location | End time             | Duration  |
|--------------------------------------|-------------|----------------|----------------------|--------------|----------|----------------------|-----------|
| DE_Cognitive_Matchmaking (Time 0:00) | Succeeded   |                | 08/28/2025, 12:01 PM |              |          | 08/28/2025, 12:01 PM | 19m 53s   |
|                                      | ✓ Succeeded | Data pipeline  | 08/28/2025, 12:01 PM |              |          | 08/28/2025, 12:20 PM | 19m 53s   |
|                                      | ✓ Succeeded | Notebook       | 08/28/2025, 10:24 AM |              |          | 08/28/2025, 10:47 AM | 23m 26s   |
|                                      | ✓ Succeeded | Notebook       | 08/28/2025, 10:17 AM |              |          | 08/28/2025, 10:24 AM | 6m 45s    |
|                                      | ✓ Succeeded | Semantic model | 08/28/2025, 9:46 AM  |              |          | 08/28/2025, 9:52 AM  | 6m 33s    |
|                                      | ✓ Succeeded | Data pipeline  | 08/28/2025, 9:00 AM  |              |          | 08/28/2025, 9:29 AM  | 29m 30s   |
|                                      | ✓ Succeeded | Semantic model | 08/28/2025, 8:31 AM  |              |          | 08/28/2025, 8:38 AM  | 7m 39s    |
|                                      | ✓ Succeeded | Data pipeline  | 08/28/2025, 8:00 AM  |              |          | 08/28/2025, 8:29 AM  | 29m 42s   |
|                                      | ✓ Succeeded | Semantic model | 08/28/2025, 6:01 AM  |              |          | 08/28/2025, 6:07 AM  | 5m 55s    |
|                                      | ✓ Succeeded | Semantic model | 08/28/2025, 6:00 AM  |              |          | 08/28/2025, 6:18 AM  | 18m 19s   |
|                                      | ✗ Failed    | Notebook       | 08/28/2025, 5:46 AM  |              |          | 08/28/2025, 6:56 AM  | 1h 9m 57s |
|                                      | ✗ Failed    | Data pipeline  | 08/28/2025, 5:46 AM  |              |          | 08/28/2025, 6:57 AM  | 1h 11m 0s |
|                                      | ✓ Succeeded | Semantic model | 08/28/2025, 5:07 AM  |              |          | 08/28/2025, 5:20 AM  | 13m 26s   |
|                                      | ✓ Succeeded | Dataflow Gen2  | 08/28/2025, 5:00 AM  |              |          | 08/28/2025, 5:06 AM  | 5m 53s    |

# How to identify what is going wrong?

## 1. Fabric Monitor tab



### Identify impact of the current outage

-  Identify running operations
  - which operations need to continue?
  - Who/what is impacted?
  - Any long running operations that we can cancel to free capacity?
-  Identify failed operations – which operations will need to be restarted?

- ✓ Part of Fabric by default
- ✗ Does not show interactive activities
- ✗ Only shows operations to which you have access
- ✗ Lacks detailed analysis

# How to identify what is going wrong?

## 2. Fabric capacity metrics app



### Detailed point-in-time analysis of throttling & rejection

- Identify which operations get throttled or rejected
- Identify which operations are causing the capacity to be full

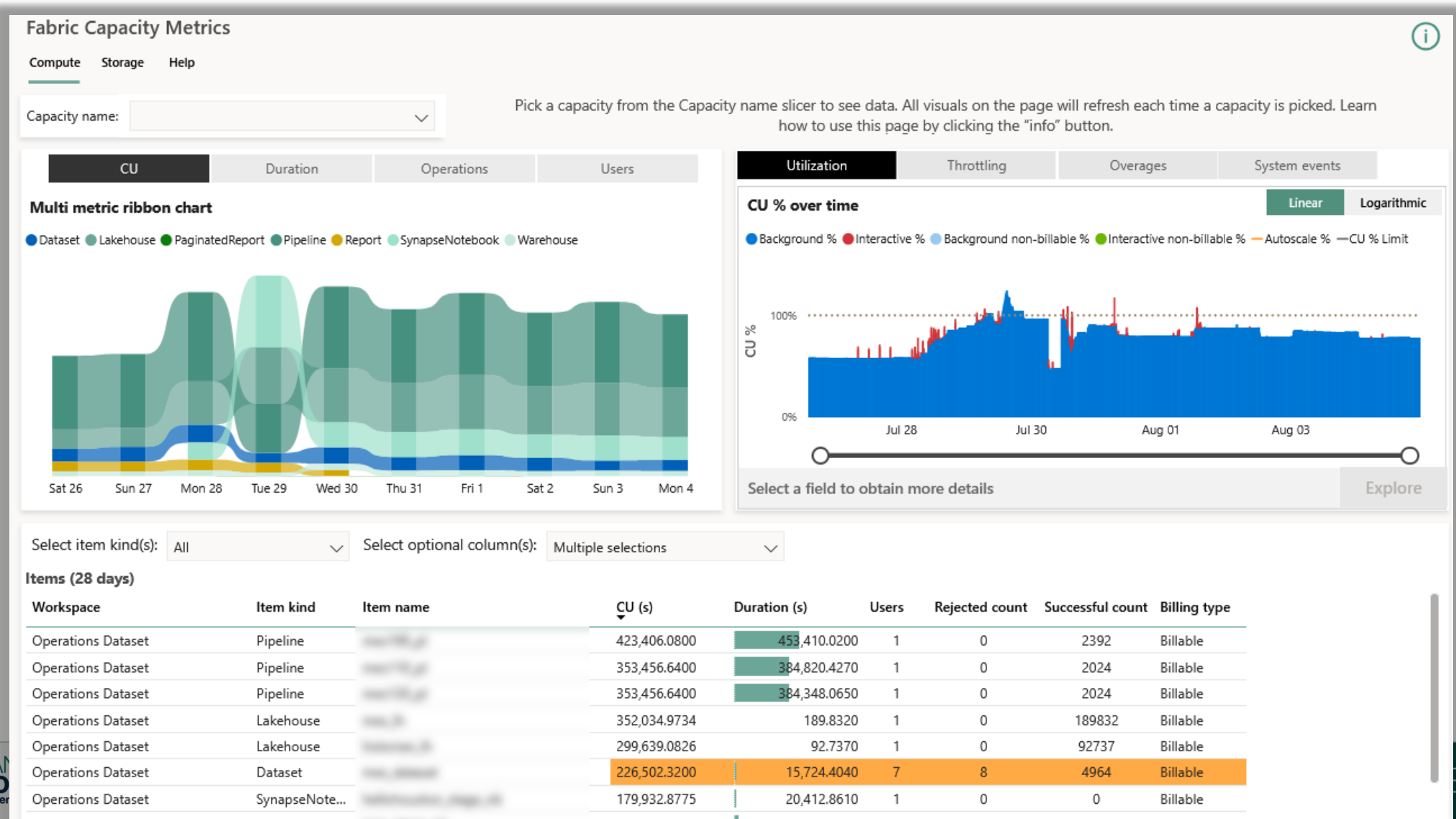
### Monitor problematic operations

Install in Power BI via Apps > Get Apps



# How to identify what is going wrong?

## 2. Fabric capacity metrics app



## Fabric Capacity Metrics

Compute Storage Help

Capacity name:

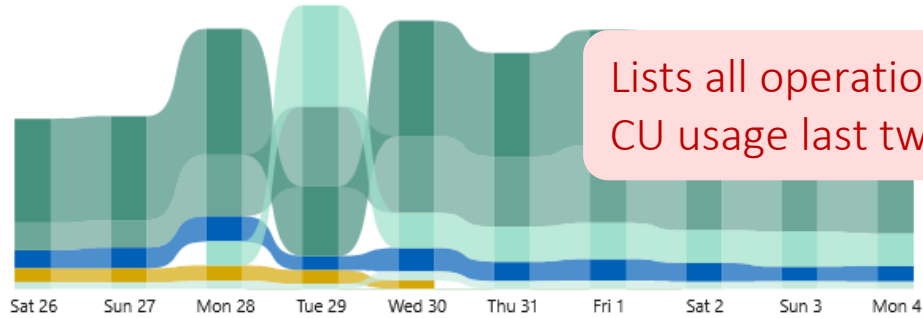
Pick a capacity from the Capacity name slicer to see data how to use this

# Overview

CU Duration Operations Users

### Multi metric ribbon chart

Dataset Lakehouse PaginatedReport Pipeline Report SynapseNotebook Warehouse

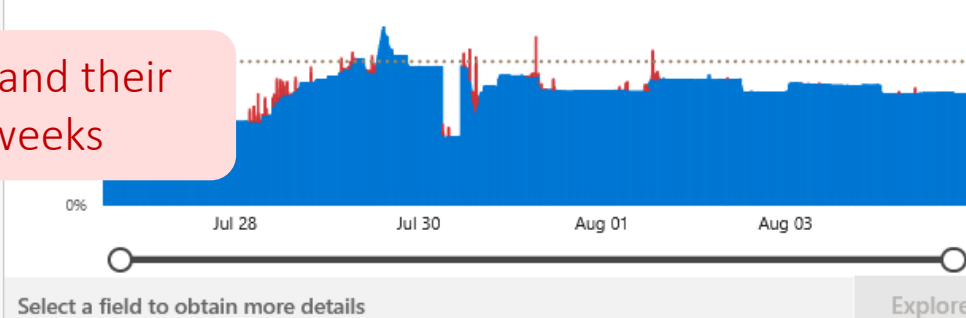


Lists all operations and their CU usage last two weeks

Utilization Throttling Overages System events

### CU % over time

Background % Interactive % Background non-billable % Interactive non-billable % Autoscale % CU % Limit



Shows which operation types take most of the CU's

Select item kind(s): All Select optional column(s): Multiple selections

### Items (28 days)

| Workspace          | Item kind      | Item name | CU (s)       | Duration (s) | Users | Rejected count | Successful count | Billing type |
|--------------------|----------------|-----------|--------------|--------------|-------|----------------|------------------|--------------|
| Operations Dataset | Pipeline       | ...       | 423,406.0800 | 453,410.0200 | 1     | 0              | 2392             | Billable     |
| Operations Dataset | Pipeline       | ...       | 353,456.6400 | 384,820.4270 | 1     | 0              | 2024             | Billable     |
| Operations Dataset | Pipeline       | ...       | 353,456.6400 | 384,348.0650 | 1     | 0              | 2024             | Billable     |
| Operations Dataset | Lakehouse      | ...       | 352,034.9734 | 189.8320     | 1     | 0              | 189832           | Billable     |
| Operations Dataset | Lakehouse      | ...       | 299,639.0826 | 92.7370      | 1     | 0              | 92737            | Billable     |
| Operations Dataset | Dataset        | ...       | 226,502.3200 | 15,724.4040  | 7     | 8              | 4964             | Billable     |
| Operations Dataset | SynapseNote... | ...       | 179,932.8775 | 20,412.8610  | 1     | 0              | 0                | Billable     |
| Operations Dataset | SynapseNote... | ...       | 120,137.5520 | 30,034.1130  | 1     | 0              | 0                | Billable     |
| Operations Dataset | Pipeline       | ...       | 103,106.8800 | 53,016.7410  | 1     | 0              | 518              | Billable     |
| Operations Dataset | Lakehouse      | ...       | 92,880.3200  | 72.9910      | 1     | 0              | 72991            | Billable     |
| Operations Dataset | SynapseNote... | ...       | 87,448.5390  | 21,861.9250  | 1     | 0              | 0                | Billable     |
| Operations Dataset | Report         | ...       | 75,600.0000  | 3,139.5350   | 1     | 0              | 189              | Billable     |
| Operations Dataset | Warehouse      | ...       | 68,227.2942  | 869,347.2570 | 4     | 0              | 19938            | Billable     |
| Operations Dataset | Warehouse      | ...       | 11,415.3464  | 7,853.3370   | 4     | 0              | 20950            | Billable     |

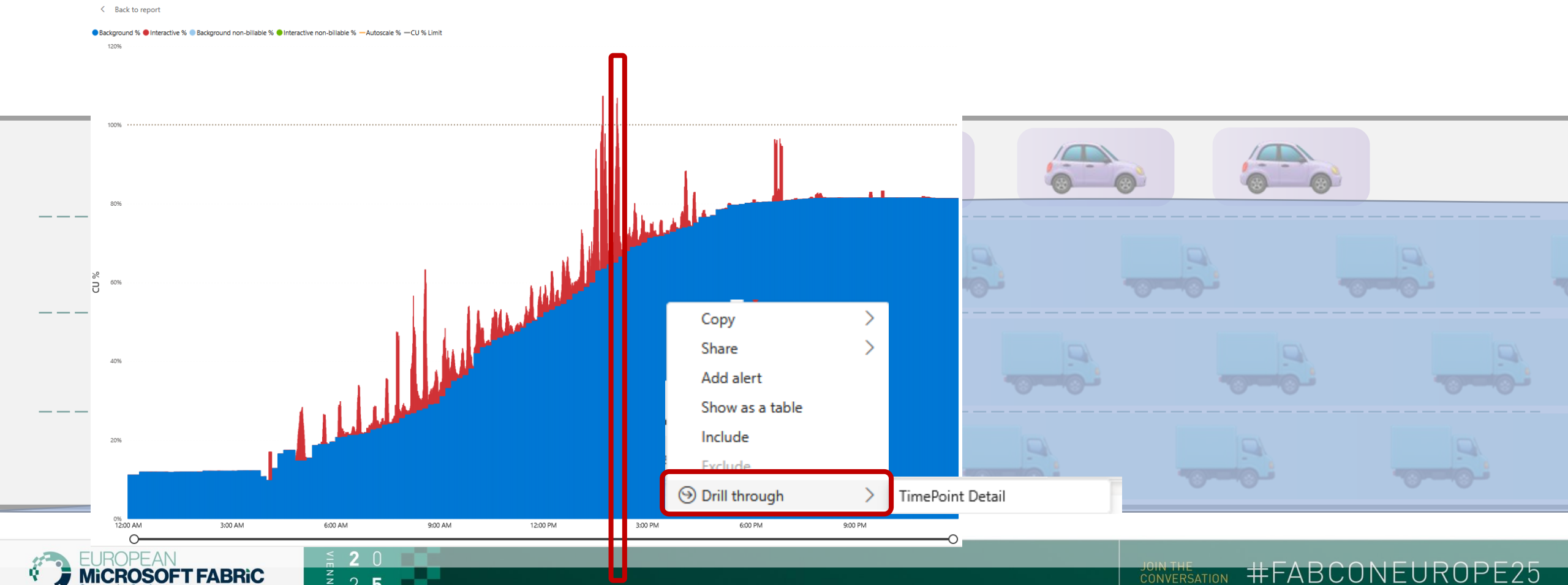
Gives an overview of evolution of the capacity load of background & interactive operations

# How to identify what is going wrong?

## 2. Fabric capacity metrics app



### Analyze the accumulated fill of the capacity



## Fabric Capacity Metrics

Start: 7/29/2025 12:06:30 PM

End: 7/29/2025 12:07:00 PM



23

Interactive operations

60K

Background operations

F4

SKU

120

CU (s)

### Interactive operations for timerange

Select optional column(s): Billing type

| Workspace          | Item kind | Item name   | Operation | Start                | End                  | Status  | User | Duration (s) | Total CU (s) | Timepoint CU (s) | Throttling (s) | % of Base capacity | Billing type |
|--------------------|-----------|-------------|-----------|----------------------|----------------------|---------|------|--------------|--------------|------------------|----------------|--------------------|--------------|
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:04:4... | 7/29/2025 12:04:4... | Success |      | 0            | 19.2480      | 1.9248           | 0              | 1.60%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:01:2... | 7/29/2025 12:01:2... | Success |      | 0            | 19.0080      | 1.9008           | 0              | 1.58%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:02:4... | 7/29/2025 12:02:4... | Success |      | 0            | 19.0080      | 1.9008           | 0              | 1.58%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:03:2... | 7/29/2025 12:03:2... | Success |      | 0            | 16.5120      | 1.6512           | 0              | 1.38%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:04:0... | 7/29/2025 12:04:0... | Success |      | 1            | 15.7440      | 1.5744           | 0              | 1.31%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:02:0... | 7/29/2025 12:02:0... | Success |      | 0            | 15.2480      | 1.5248           | 0              | 1.27%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:04:0... | 7/29/2025 12:04:0... | Success |      | 0            | 5.7440       | 0.5744           | 0              | 0.48%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:04:0... | 7/29/2025 12:04:0... | Success |      | 0            | 0.7520       | 0.0752           | 0              | 0.06%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:01:2... | 7/29/2025 12:01:2... | Success |      | 0            | 0.5120       | 0.0512           | 0              | 0.04%              | Billable     |
| Operations Dataset | Dataset   | mes_dataset | Query     | 7/29/2025 12:04:4... | 7/29/2025 12:04:4... | Success |      | 0            | 0.5120       | 0.0512           | 0              | 0.04%              | Billable     |
| Total              |           |             |           |                      |                      |         |      | 1            | 112.7840     | 11.2784          | 0              | 9.40%              |              |

### Background operations for timerange

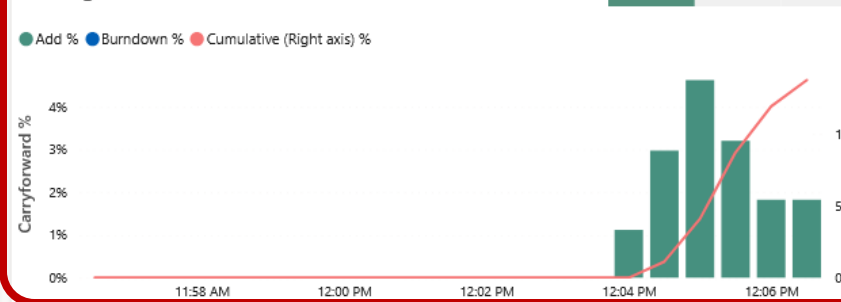
Select optional column(s): Billing type

| Workspace          | Item kind  | Item name   | Operation       | Start               | End                 | Status    | User | Duration (s) | Total CU (s) | Timepoint CU (s) | Throttling (s) | % of Base capacity | Billing type |
|--------------------|------------|-------------|-----------------|---------------------|---------------------|-----------|------|--------------|--------------|------------------|----------------|--------------------|--------------|
| Operations Dataset | Synapse... |             | Notebook Run    | 7/29/2025 6:37:0... | 7/29/2025 7:25:0... | Cancelled |      | 2,881        | 24,251.8290  | 8.4208           | 0              | 7.02%              | Billable     |
| Operations Dataset | Synapse... |             | Notebook Pip... | 7/28/2025 5:03:1... | 7/28/2025 5:22:1... | Cancelled |      | 1,133        | 10,354.2050  | 3.5952           | 0              | 3.00%              | Billable     |
| Operations Dataset | Synapse... |             | Notebook Run    | 7/29/2025 7:40:2... | 7/29/2025 7:49:2... | Cancelled |      | 537          | 5,325.8440   | 1.8493           | 0              | 1.54%              | Billable     |
| Operations Dataset | Dataset    | mes_dataset | Dataset On-D... | 7/29/2025 6:12:1... | 7/29/2025 6:14:5... | Success   |      | 163          | 3,185.7440   | 1.1062           | 0              | 0.92%              | Billable     |
| Operations Dataset | Dataset    | mes_dataset | Dataset On-D... | 7/28/2025 3:29:0... | 7/28/2025 3:31:1... | Success   |      | 123          | 2,773.0080   | 0.9629           | 0              | 0.80%              | Billable     |
| Operations Dataset | Dataset    | mes_dataset | Dataset On-D... | 7/28/2025 12:23:... | 7/28/2025 12:25:... | Success   |      | 125          | 2,770.7520   | 0.9621           | 0              | 0.80%              | Billable     |
| Operations Dataset | Dataset    | mes_dataset | Dataset On-D... | 7/29/2025 11:26:... | 7/29/2025 11:28:... | Success   |      | 128          | 2,503.0080   | 0.8691           | 0              | 0.72%              | Billable     |
| Operations Dataset | Dataset    | mes_dataset | Dataset On-D... | 7/28/2025 4:25:1... | 7/28/2025 4:27:0... | Success   |      | 119          | 2,488.0000   | 0.8639           | 0              | 0.72%              | Billable     |
| Operations Dataset | Dataset    | mes_dataset | Dataset On-D... | 7/29/2025 4:25:0... | 7/29/2025 4:26:5... | Success   |      | 106          | 2,392.4960   | 0.8307           | 0              | 0.69%              | Billable     |
| Total              |            |             |                 |                     |                     |           |      | 235,404      | 319,324.2... | 110.8765         | 0              | 92.40%             |              |

### Burndown table for timerange

| Experience       | Add % | Burndown % | Cumulative % | Minutes to burndown |
|------------------|-------|------------|--------------|---------------------|
| DI               | 0.63% | 0.00%      | 4.70%        | 0.02                |
| Iake             | 0.43% | 0.00%      | 3.22%        | 0.02                |
| SparkCore        | 0.35% | 0.00%      | 2.58%        | 0.01                |
| AS               | 0.29% | 0.00%      | 2.33%        | 0.01                |
| ScreenshotEngine | 0.08% | 0.00%      | 0.62%        | 0.00                |
| DMS              | 0.04% | 0.00%      | 0.27%        | 0.00                |
| RsRdlEngine      | 0.00% | 0.00%      | 0.01%        | 0.00                |
| Total            | 1.82% | 0.00%      | 13.72%       | 0.07                |

### Overages



# TimePoint Detail

When drilling through on timeline

Shows the percentage these operations take at that point in time (30 seconds)

$$9.4 \% + 92.4 \% = 101.8\%$$

Exceeding the capacity results in overages. If the overages can not be « resolved » within 10 mins, throttling starts

Start: 8/16/2025 6:43:00 PM

End: 8/16/2025 6:43:30 PM

CU % Limit

CU %

(Blank)

44K

F4

120

Interactive operations

Background operations

SKU

CU (s)

Interactive operations for timerange

Select optional column(s): Billing type

| Workspace | Item kind | Item name | Operation | Start | End | Status | User | Duration (s) | Total CU (s) | Timepoint CU (s) | Throttling (s) | % of Base capacity | Billing type |
|-----------|-----------|-----------|-----------|-------|-----|--------|------|--------------|--------------|------------------|----------------|--------------------|--------------|
|           |           |           |           |       |     |        |      |              |              |                  |                |                    |              |

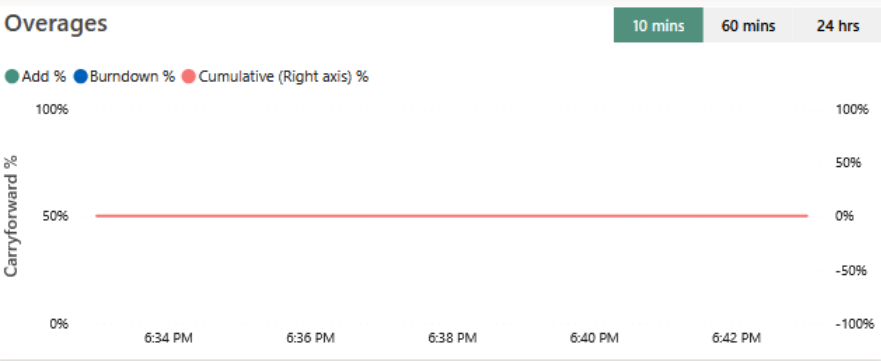
Background operations for timerange

Select optional column(s): Billing type

| Workspace          | Item kind | Item name | Operation        | Start               | End                  | Status  | User               | Duration (s) | Total CU (s) | Timepoint CU (s) | Throttling (s) | % of Base capacity | Billing type |
|--------------------|-----------|-----------|------------------|---------------------|----------------------|---------|--------------------|--------------|--------------|------------------|----------------|--------------------|--------------|
| Operations Dataset | Wareho... |           | OneLake Othe...  | 8/15/2025 9:00:0... | 8/15/2025 9:00:00 PM | Success |                    | 0            | 0.0104       | 0.0000           | 0              | 0.00%              | Billable     |
| Operations Dataset | Pipeline  |           | ActivityRun      | 8/15/2025 9:00:1... | 8/15/2025 9:12:04 PM | Success | 1bdb6a12-f4fb-4... | 712          | 20.1600      | 0.0070           | 0              | 0.01%              | Billable     |
| Operations Dataset | Pipeline  |           | ActivityRun      | 8/15/2025 9:00:1... | 8/15/2025 9:26:02 PM | Success | 1bdb6a12-f4fb-4... | 1,550        | 20.1600      | 0.0070           | 0              | 0.01%              | Billable     |
| Operations Dataset | Wareho... |           | OneLake Othe...  | 8/15/2025 9:00:3... | 8/15/2025 9:00:30 PM | Success |                    | 0            | 0.0624       | 0.0000           | 0              | 0.00%              | Billable     |
| Operations Dataset | Wareho... |           | OneLake Read...  | 8/15/2025 9:00:3... | 8/15/2025 9:00:30 PM | Success |                    | 0            | 1.4664       | 0.0005           | 0              | 0.00%              | Billable     |
| Operations Dataset | Pipeline  |           | ActivityRun      | 8/15/2025 9:00:3... | 8/15/2025 9:00:53 PM | Success | 1bdb6a12-f4fb-4... | 17           | 20.1600      | 0.0070           | 0              | 0.01%              | Billable     |
| Operations Dataset | Pipeline  |           | ActivityRun      | 8/15/2025 9:00:3... | 8/15/2025 9:00:54 PM | Success | 1bdb6a12-f4fb-4... | 17           | 20.1600      | 0.0070           | 0              | 0.01%              | Billable     |
| Operations Dataset | Pipeline  |           | ActivityRun      | 8/15/2025 9:00:3... | 8/15/2025 9:09:48 PM | Success | 1bdb6a12-f4fb-4... | 1,653        | 60.4800      | 0.0210           | 0              | 0.02%              | Billable     |
| Operations Dataset | Wareho... |           | SQL Endpoint ... | 8/15/2025 9:00:4... | 8/15/2025 9:00:42 PM | Success | System             | 0            | 0.9800       | 0.0003           | 0              | 0.00%              | Billable     |
| Total              |           |           |                  |                     |                      |         |                    | 297,506      | 208,700.5... | 72.4655          | 0              | 60.39%             |              |

Burndown table for timerange

| Experience  | Add % | Burndown % | Cumulative % | Minutes to burndown |
|-------------|-------|------------|--------------|---------------------|
| AS          | 0.00% | 0.00%      | 0.00%        | 0.00                |
| DI          | 0.00% | 0.00%      | 0.00%        | 0.00                |
| DMS         | 0.00% | 0.00%      | 0.00%        | 0.00                |
| lake        | 0.00% | 0.00%      | 0.00%        | 0.00                |
| RsRdlEngine | 0.00% | 0.00%      | 0.00%        | 0.00                |
| SparkCore   | 0.00% | 0.00%      | 0.00%        | 0.00                |
| Total       | 0.00% | 0.00%      | 0.00%        | 0.00                |



# TimePoint Detail

When drilling through on timeline

Background operations are not necessarily running operations but *smoothed* usage.

Operations from Aug 15th – 9PM Are still being paid back at Aug 16th – 6:43PM

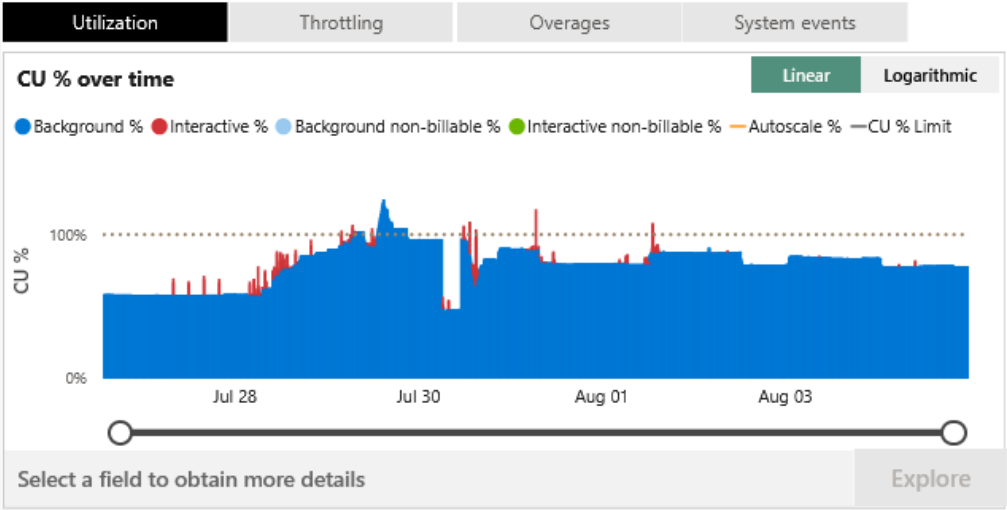
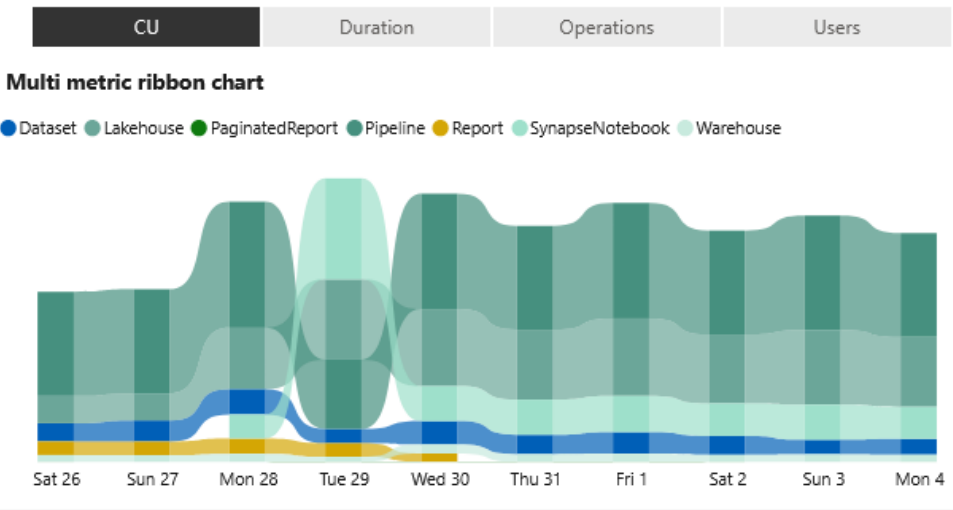
Fabric Capacity Metrics

Compute Storage Help



Capacity name:

Pick a capacity from the Capacity name slicer to see data. All visuals on the page will refresh each time a capacity is picked. Learn how to use this page by clicking the "info" button.



Select item kind(s): All Select optional column(s): Multiple selections

Items (28 days)

| Workspace          | Item kind      | Item name | CU (s)       | Duration (s) | Users | Rejected count | Successful count | Billing type |
|--------------------|----------------|-----------|--------------|--------------|-------|----------------|------------------|--------------|
| Operations Dataset | Pipeline       |           | 423,406.0800 | 453,410.0200 | 1     | 0              | 2392             | Billable     |
| Operations Dataset | Pipeline       |           | 353,456.6400 | 384,820.4270 | 1     | 0              | 2024             | Billable     |
| Operations Dataset | Pipeline       |           | 353,456.6400 | 384,348.0650 | 1     | 0              | 2024             | Billable     |
| Operations Dataset | Lakehouse      |           | 352,034.9734 | 189.8320     | 1     | 0              | 189832           | Billable     |
| Operations Dataset | Lakehouse      |           | 299,639.0826 | 92.7370      | 1     | 0              | 92737            | Billable     |
| Operations Dataset | Dataset        |           | 226,502.3200 | 15,724.4040  | 7     | 8              | 4964             | Billable     |
| Operations Dataset | SynapseNote... |           | 179,932.8775 | 20,412.8610  | 1     | 0              | 0                | Billable     |
| Operations Dataset | SynapseNote... |           | 120,137.5520 | 30,034.1130  | 1     | 0              | 0                | Billable     |
| Operations Dataset | Pipeline       |           | 103,106.8800 | 53,016.7410  | 1     | 0              | 518              | Billable     |
| Operations Dataset | Lakehouse      |           | 92,880.3200  | 72.9910      | 1     | 0              | 72991            | Billable     |
| Operations Dataset | SynapseNote... |           | 87,448.5390  | 21,861.9250  | 1     | 0              | 0                | Billable     |
| Operations Dataset | Report         |           | 75,600.0000  | 3,139.5350   | 1     | 0              | 189              | Billable     |
| Operations Dataset | Warehouse      |           | 68,227.2942  | 869,347.2570 | 4     | 0              | 19938            | Billable     |
| Operations Dataset | Warehouse      |           | 11,415.3464  | 7,853.3370   | 4     | 0              | 20950            | Billable     |

Overview

# Item performance

| Workspace | Item kind       | Item name | CU (s)          | Duration (s)    | Users | Rejected count | Billing type |
|-----------|-----------------|-----------|-----------------|-----------------|-------|----------------|--------------|
|           | Dataset         |           | 3,416,252.9504  | 71,512.9850     | 40    | 0              | Billable     |
|           | EventStream     |           | 2,779,502.2454  | 4,644,000.0000  | 1     | 0              | Billable     |
|           | Dataset         |           | 2,629,020.0000  | 53,083.1020     | 12    | 0              | Billable     |
|           | Dataset         |           | 2,427,876.0736  | 110,982.3480    | 8     | 0              | Billable     |
|           | KustoEventHouse |           | 2,421,360.0000  | 1,076,160.0000  | 1     | 0              | Billable     |
|           | Dataflow        |           | 2,227,092.6080  | 7,133.4500      | 1     | 0              | Billable     |
|           | Dataset         |           | 2,048,595.0176  | 62,015.8850     | 6     | 0              | Billable     |
|           | Dataflow        |           | 1,606,591.2320  | 53,646.0690     | 1     | 0              | Billable     |
|           | Dataset         |           | 1,411,460.2016  | 33,534.1390     | 32    | 0              | Billable     |
|           | Dataset         |           | 1,337,825.4048  | 50,669.1580     | 10    | 0              | Billable     |
|           | Dataset         |           | 1,249,125.2576  | 23,145.3560     |       |                |              |
|           | Dataflow        |           | 1,222,415.9680  | 100,616.3570    |       |                |              |
|           | SynapseNotebook |           | 1,131,452.7451  | 36,950.9360     |       |                |              |
|           | Dataflow        |           | 888,048.0320    | 20,545.1030     |       |                |              |
|           | Dataset         |           | 884,298.4192    | 56,896.5310     |       |                |              |
|           | Dataflow        |           | 856,329.0240    | 78,744.2380     |       |                |              |
|           | Dataset         |           | 717,641.5712    | 33,805.1950     |       |                |              |
|           | SynapseNotebook |           | 653,107.4994    | 49,812.7630     |       |                |              |
|           | SynapseNotebook |           | 566,256.6660    | 50,117.6230     |       |                |              |
|           | Dataset         |           | 544,425.5776    | 11,023.3580     |       |                |              |
|           | Dataset         |           | 517,685.0816    | 33,948.9230     |       |                |              |
|           | Dataflow        |           | 509,950.7040    | 41,721.2630     |       |                |              |
|           | Dataset         |           | 506,083.2288    | 16,693.4420     | 2     |                |              |
|           | Dataset         |           | 472,687.5808    | 7,853.0770      |       |                |              |
|           | Dataset         |           | 447,859.7344    | 14,789.2440     |       |                |              |
|           | Dataset         |           | 442,175.0336    | 12,406.9300     |       |                |              |
|           | Dataset         |           | 413,437.6320    | 16,039.9100     |       |                |              |
|           | Dataset         |           | 410,562.7808    | 16,359.6450     |       |                |              |
|           | Dataset         |           | 395,988.1280    | 10,077.0340     | 2     |                |              |
|           | Dataflow        |           | 384,625.7280    | 25,757.5940     |       |                |              |
|           | Dataflow        |           | 380,248.1600    | 69,616.3180     |       |                |              |
|           | Dataset         |           | 356,997.3024    | 19,913.0480     | 7     | 0              | Billable     |
|           | Dataset         |           | 356,443.3888    | 31,805.2890     | 11    | 0              | Billable     |
|           |                 |           | 48,050,915.8210 | 17,821,574.8000 | 40    | 0              |              |

Compare CU's and see which operations have high usage

Colour gives indication of how performance is deteriorating (but is not found to be very accurate)

# How to identify what is going wrong?

## 2. Fabric capacity metrics app

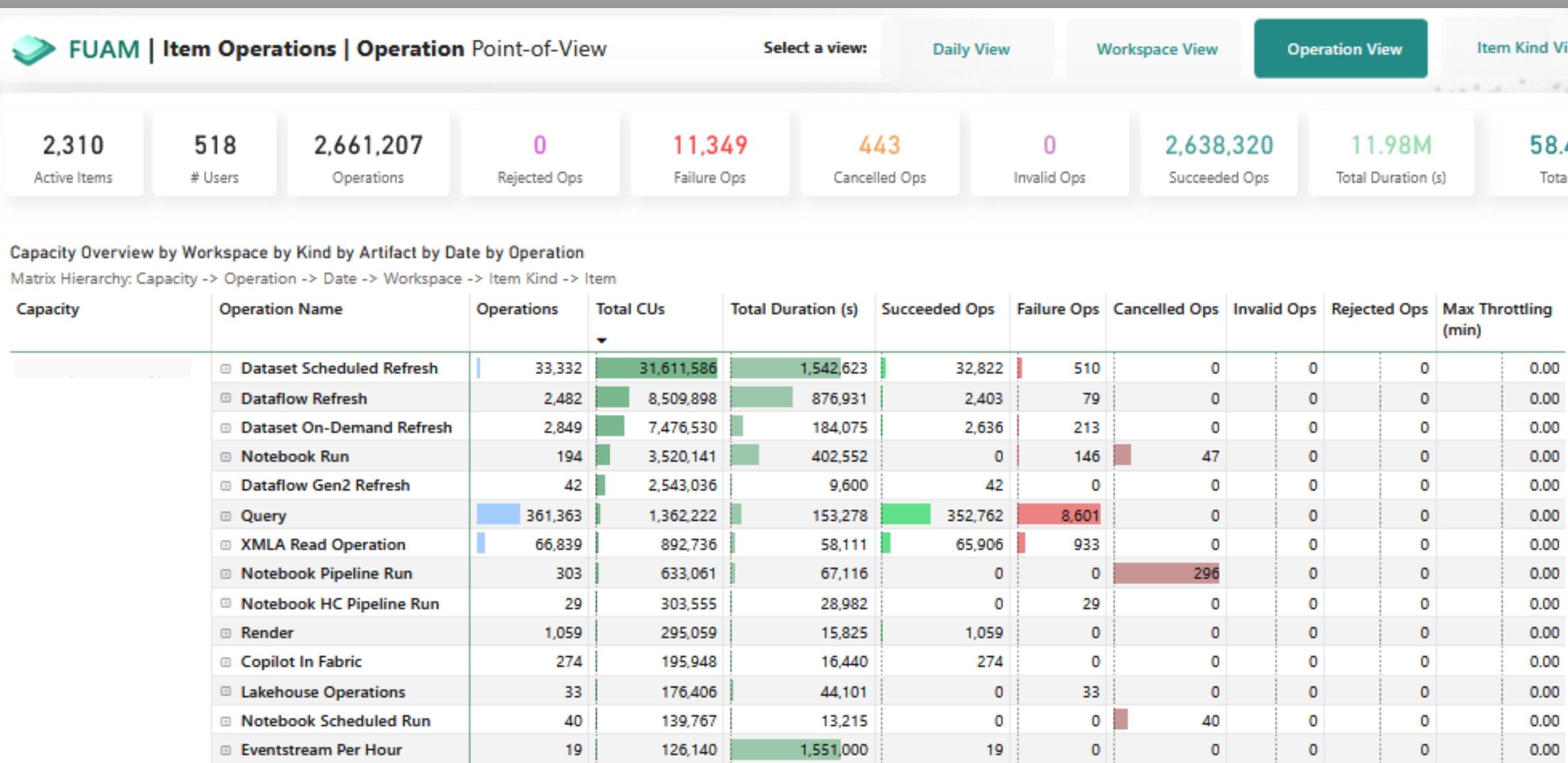


### Things to take into account

- Evaluate CU usage based on frequency of the operation
- Remember background vs. interactive activities smoothing
- You can only run the report capacity per capacity

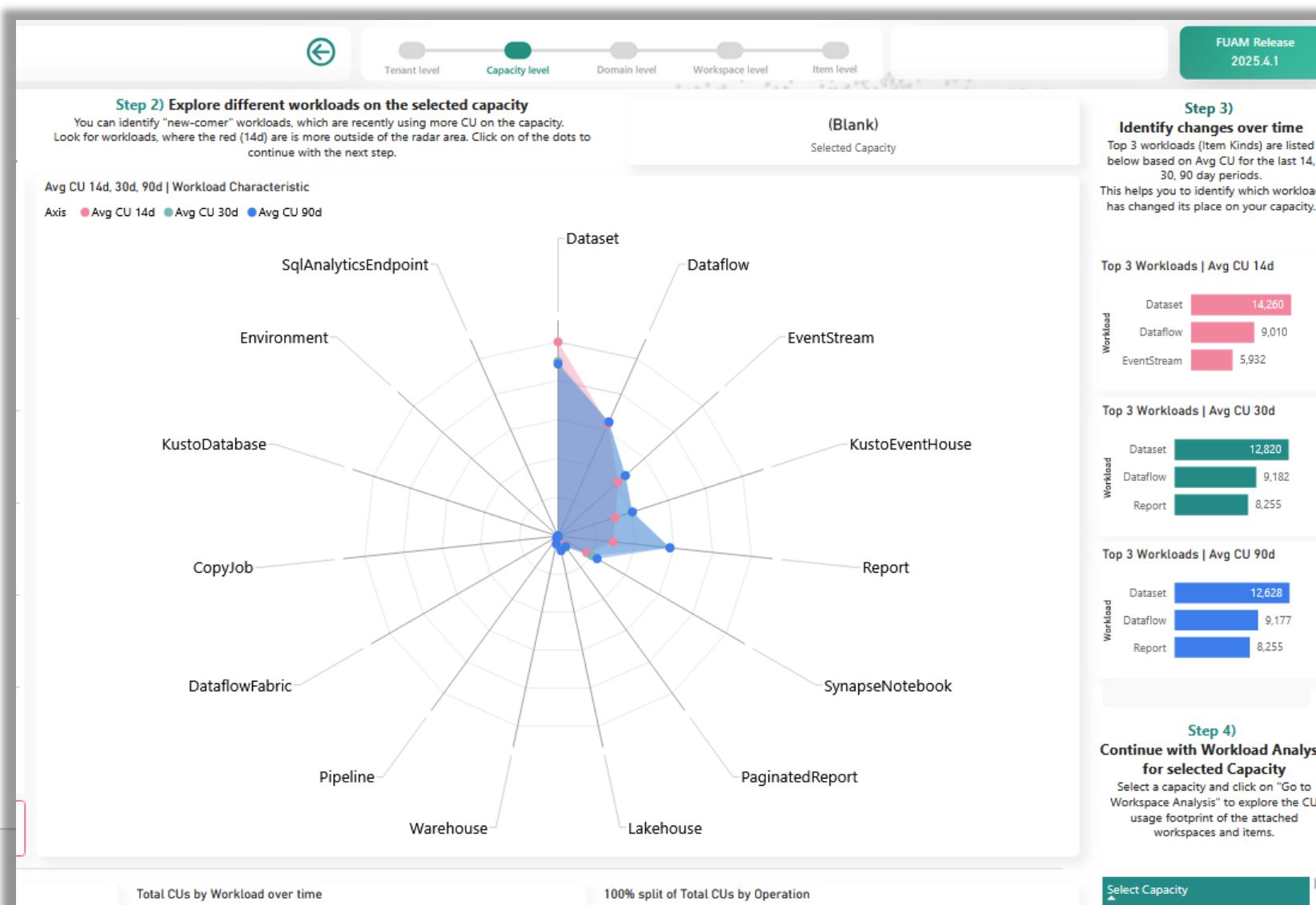
# How to identify what is going wrong?

## 3. FUAM



# How to identify what is going wrong?

## 3. FUAM – workload characteristics



# How to identify what is going wrong?

## 3. FUAM – Outlier Analysis



 **FUAM | Item Outlier Analysis**

←

Tenant level

Capacity level

Domain level

Workspace level

Item level

Selected Capacity

(Blank)  
Selected Date

(Blank)  
Max TimePoint CU %

(Blank)  
Avg TimePoint CU %

(Blank)  
Throttling Count

Item Outlier Analysis

| Date       | Workspace Name | Item Kind       | Item Name | Total CUs | Daily CU vs 14d Avg CU | Outlier Factor |
|------------|----------------|-----------------|-----------|-----------|------------------------|----------------|
| 2025-07-25 |                | DataflowFabric  |           | 1,228     | 1,193.14               |                |
| 2025-07-25 |                | Lakehouse       |           | 11,341    | 10,948.15              |                |
| 2025-07-31 |                | Pipeline        |           | 26,099    | 24,357.75              |                |
| 2025-07-30 |                | Dataset         |           | 14,634    | 13,617.28              |                |
| 2025-07-22 |                | SynapseNotebook |           | 57,788    | 53,328.40              |                |
| 2025-07-22 |                | SynapseNotebook |           | 68,125    | 62,780.11              |                |
| 2025-07-22 |                | Warehouse       |           | 2,538     | 2,316.28               |                |
| 2025-07-25 |                | Lakehouse       |           | 11,287    | 10,226.17              |                |
| 2025-07-23 |                | Dataset         |           | 9,850     | 8,913.86               |                |
| 2025-07-25 |                | Lakehouse       |           | 19,782    | 17,572.84              |                |
| 2025-07-23 |                | Dataset         |           | 195       | 171.10                 |                |
| 2025-07-25 |                | Warehouse       |           | 111       | 97.43                  |                |
| 2025-07-23 |                | Dataset         |           | 26,347    | 23,020.29              |                |
| 2025-07-28 |                | Lakehouse       |           | 8,154     | 7,095.81               |                |
| 2025-08-04 |                | Dataset         |           | 174       | 149.37                 |                |
| 2025-08-03 |                | Dataset         |           | 589       | 505.29                 |                |
| 2025-07-30 |                | Dataset         |           | 19,456    | 16,615.61              |                |

**Step 3) Check recurring outlier items**  
In this step, take a look on the table.  
Red marked items are items which have a 'bad-actor' behavior in the last 14 days.

**Daily CU vs 14d Avg CU**  
This metric matches the total daily Average CU with the daily average CU for the last 14 days of the given item.  
FUAM uses it to visualize whenever the item is using more CU in the last 14 days as in its entire lifecycle.

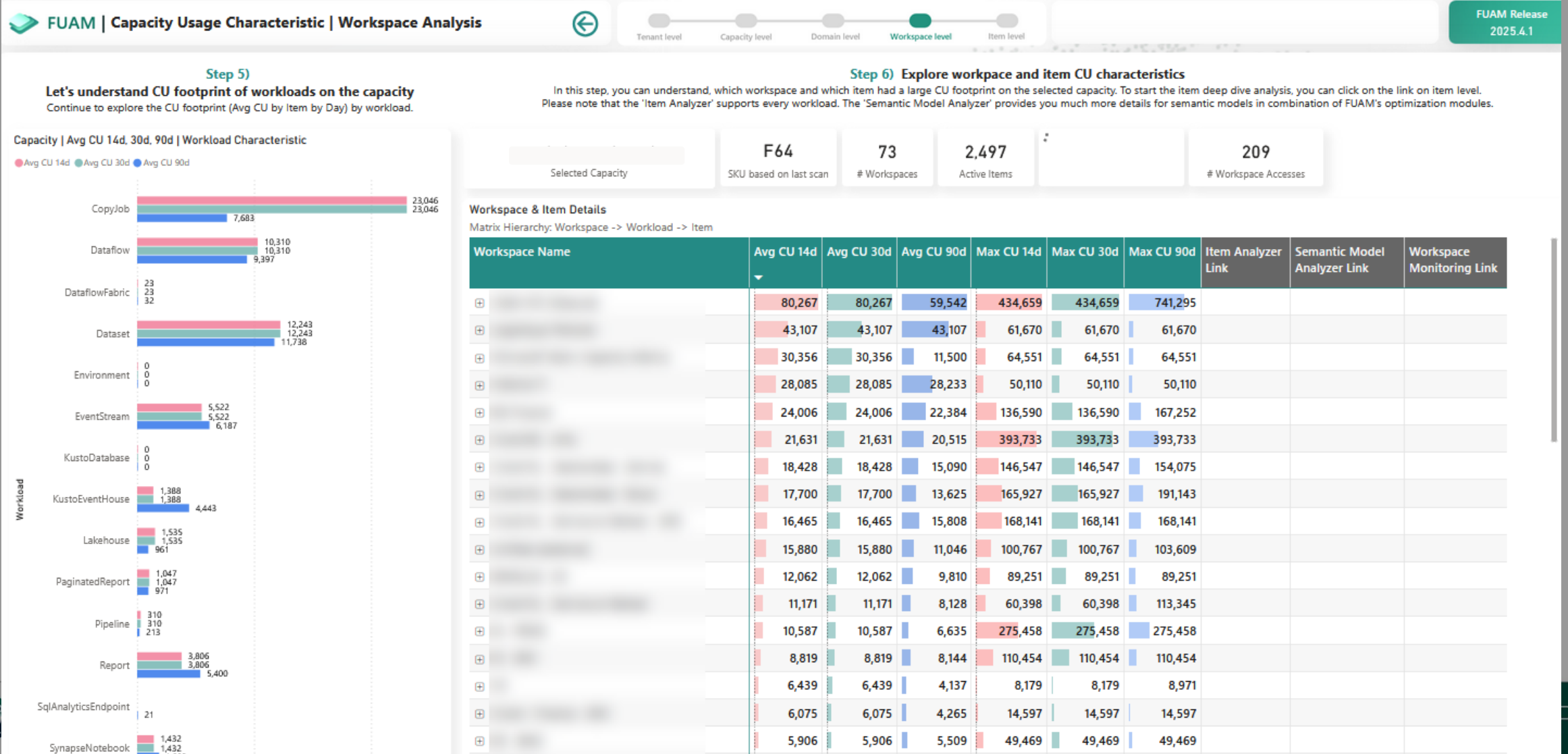
**Outlier Factor**  
Every F-SKU and every workload has its own character. The Total CU and the percentage of the capacity utilization, are key metrics, however it doesn't represents if an item is an outlier. For that reason, FUAM calculates a Outlier Factor based on standard variance.  
  
The more filled the dice symbol is, the higher the outlier factor.  
We recommend analyzing such items in the next steps.

**Step 4) Select an item and continue with in-depth analysis**  
FUAM's item analyzer provides you a unique holistic view about the selected item.

[Go to Item Analyzer](#)

# How to identify what is going wrong?

## 3. FUAM – workspace analysis



# How to identify what is going wrong?

## 3. FUAM



### Detailed overview across all capacities about expensive and troublesome operations

- Helps you prevent issues in the future and helps with smarter capacity planning
- Helps you analyze semantic model and query performance based on historical information

Install via [Github](#) (full Fabric solution)

# How to identify what goes wrong?

|              |  <b>Monitor tab</b>      |  <b>Capacity Metrics App</b> |  <b>FUAM</b> |
|--------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Installation | Already installed                                                                                         | Power BI App (Get Apps)<br>Easy                                                                                 | Workspace deployment<br>Requires some knowledge and a capacity                                  |
| Main purpose | Assess current impact of capacity issues                                                                  | Assess cause of capacity issues<br>Monitor capacity                                                             | Analyze & optimize to avoid issues in the future                                                |
| Pro          | Built in                                                                                                  | Very detailed analysis<br>Shows smoothing & overage mechanics                                                   | Cross-capacity analysis<br>Shows bottlenecks within models                                      |
| Con          | Does not show interactive activities<br>Lacks detailed analysis<br>You only see things you have access to | One capacity at the time<br>Limited history                                                                     | More complex installation                                                                       |

# How to identify what goes wrong?

- Pitfalls:
  - Looking at the wrong capacity 🧑
  - Identifying the running activities as the issue (remember smoothing)
  - Having the monitoring reports run on the overrun capacity

# Typical tasks where you can start looking

- DataFlow Gen2 (\*)
- Inefficient semantic model refresh
  - Calculated columns
  - Pivot
  - ...
- Paginated reports
- Complex user queries
  - Inefficient DAX
  - Analyze in Excel
- EventHouse
- CoPilot

(\* Dataflow Gen2 improvements were announced that will reduce CU usage)

# Typical tasks where you can start looking

## How to solve?

- Move workload to “cheaper” options
  - Dataflow gen2 → notebooks
  - Semantic model operations → transform in Warehouse or Lakehouse
  - Paginated reports → Power BI dashboards
- DAX optimisation:
  - Best Practice analyzer (in tabular editor or via semantic link labs)
  - If possible, move calculations towards Warehouse or lakehouse tables
- CoPilot
  - Leverage CoPilot capacities

# How to recover?



# Recovering a throttled capacity



1.  Wait
2.  Pause & resume
3.  Scale
4.  Switch capacities

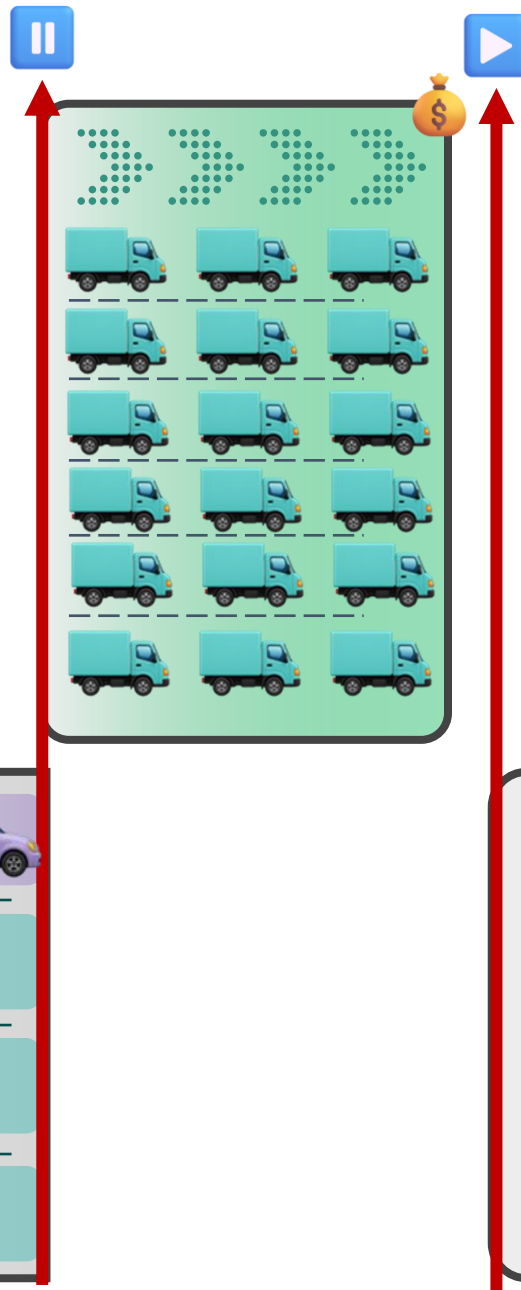
# Recovering a throttled capacity

## 1. Wait 🕒

= waiting in the traffic jam for the highway to clear up

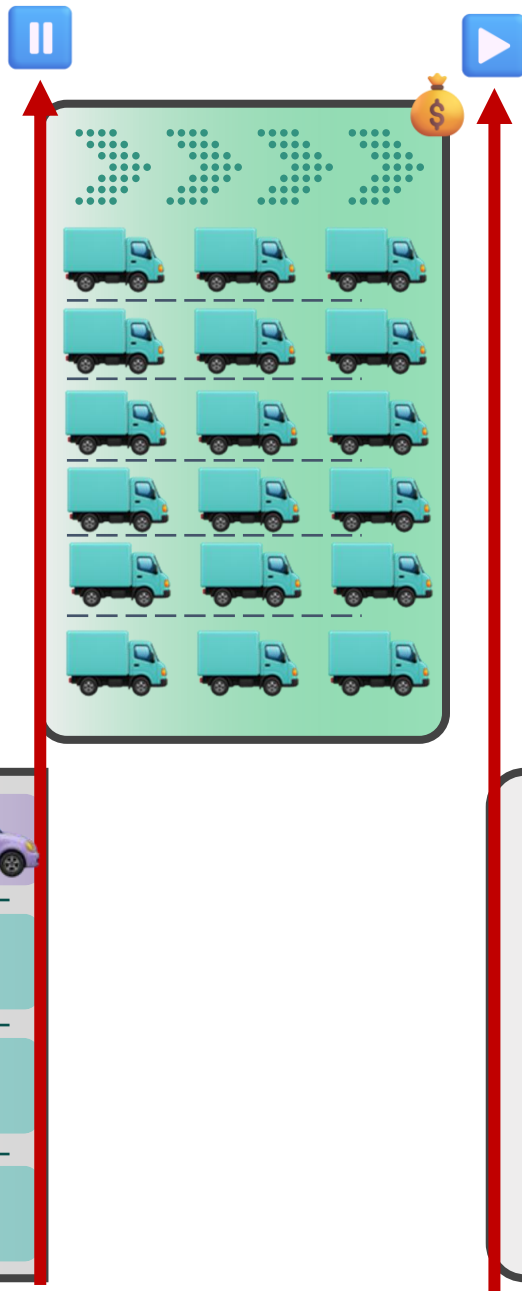
- Only possible if there's no (urgent) business impact
- Can work if there's a temporary peak of interactive operations
  - Interactive operations tend to finalize quickly
- Can take up to 24 hours if background operations are the root cause





## 2. Pause & resume

Similar to getting a temporary, superfast, tolled highway where all your “smoothed” operations are resolved immediately to clear the way.

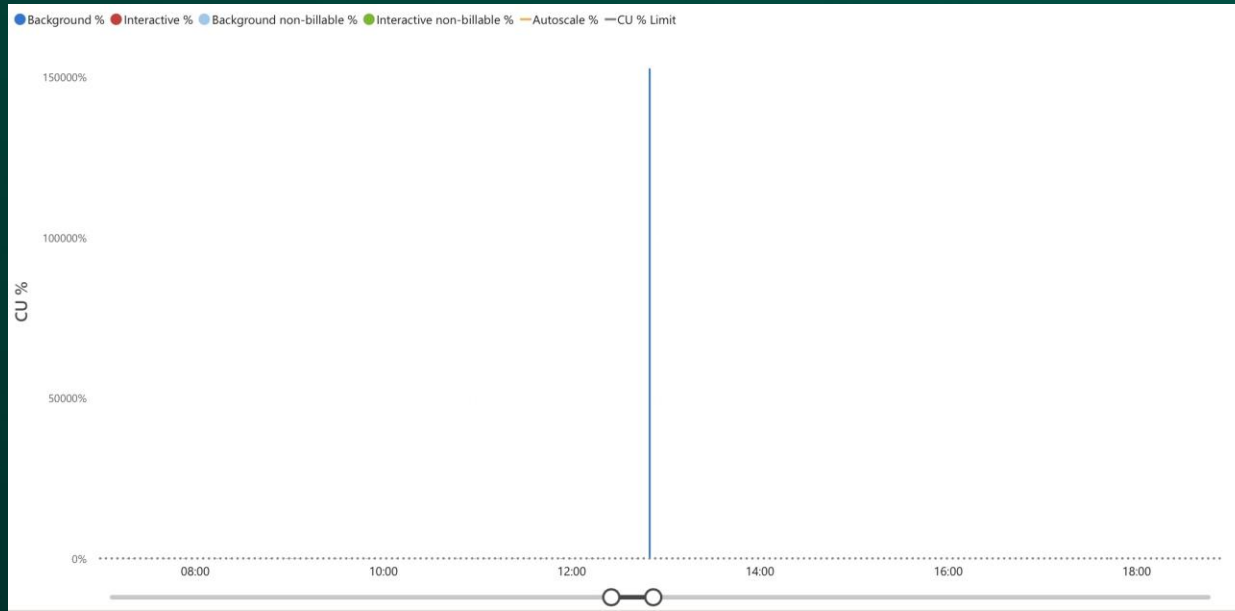


## 2. Pause & resume

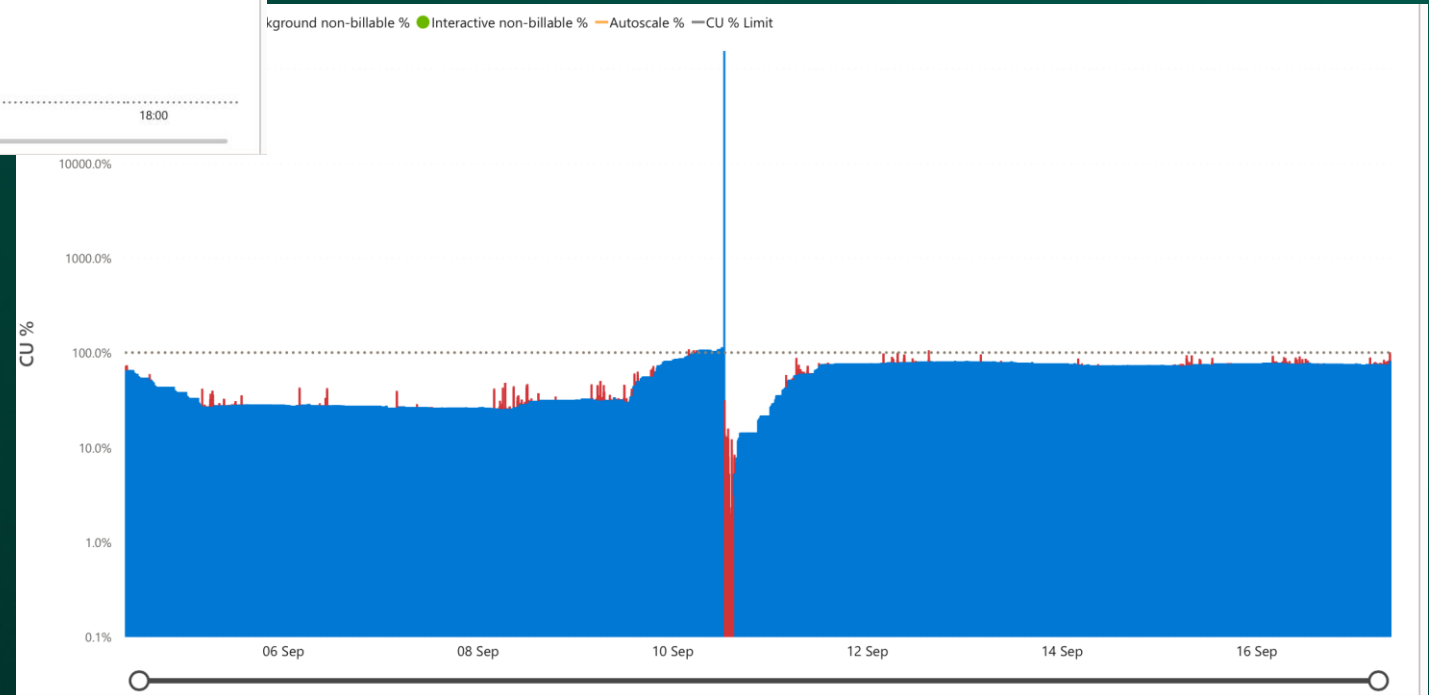
 The toll you pay is billed at pay-as-you-go pricing.

## 2. Pause & resume

Bill Shock



Linear Scale



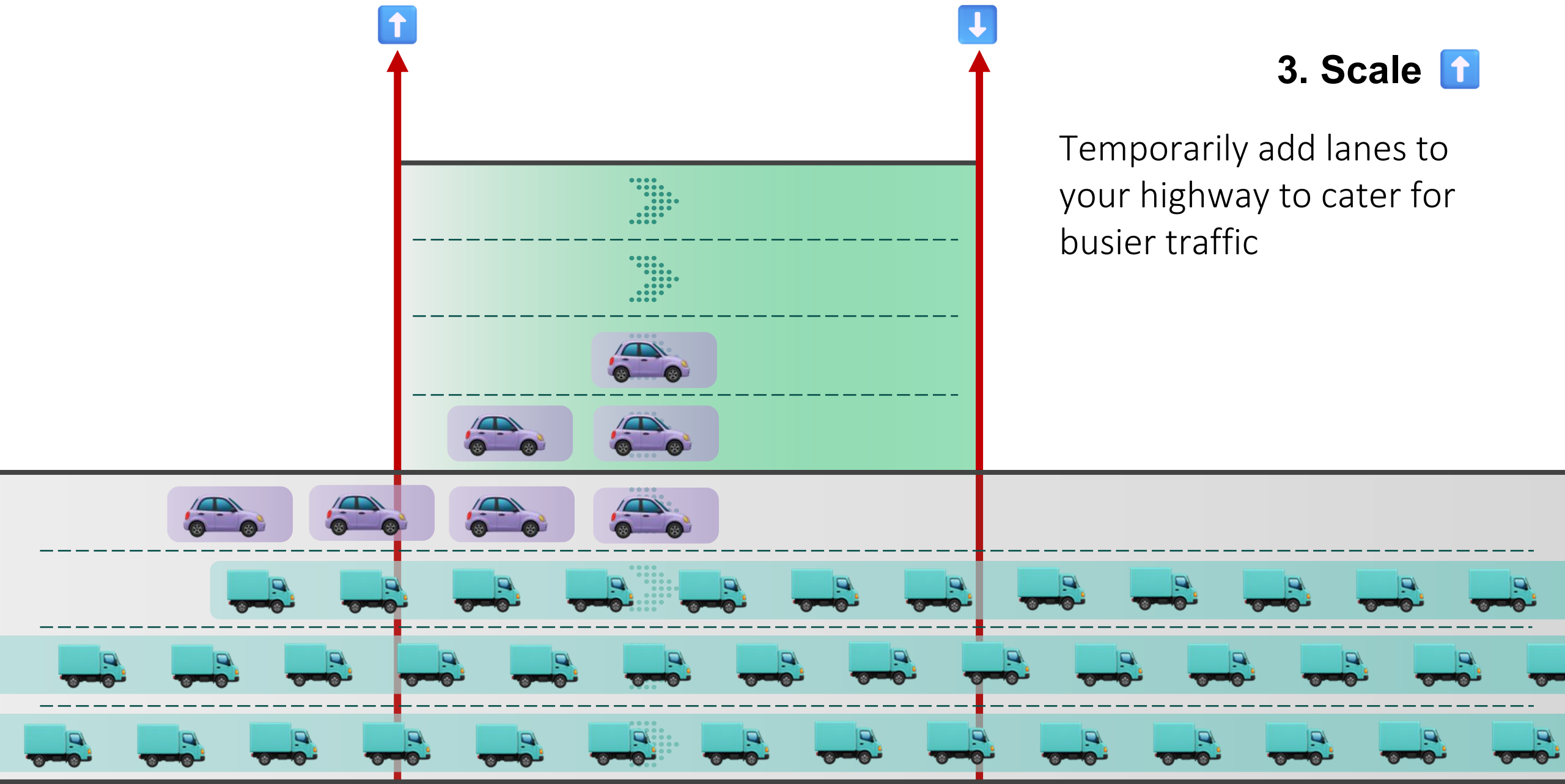
# Recovering a throttled capacity

## 2. Pause & resume

- Immediately pay for the “overage” and start with a clean capacity
- Running jobs are cancelled
- Make sure you know what caused the overrun so you can restart “light” and avoid the issue coming back
  - E.g. an inefficient semantic model refresh needs to be paused & looked after first
- Causes “bill shock” 

### 3. Scale

Temporarily add lanes to your highway to cater for busier traffic



# Recovering a throttled capacity

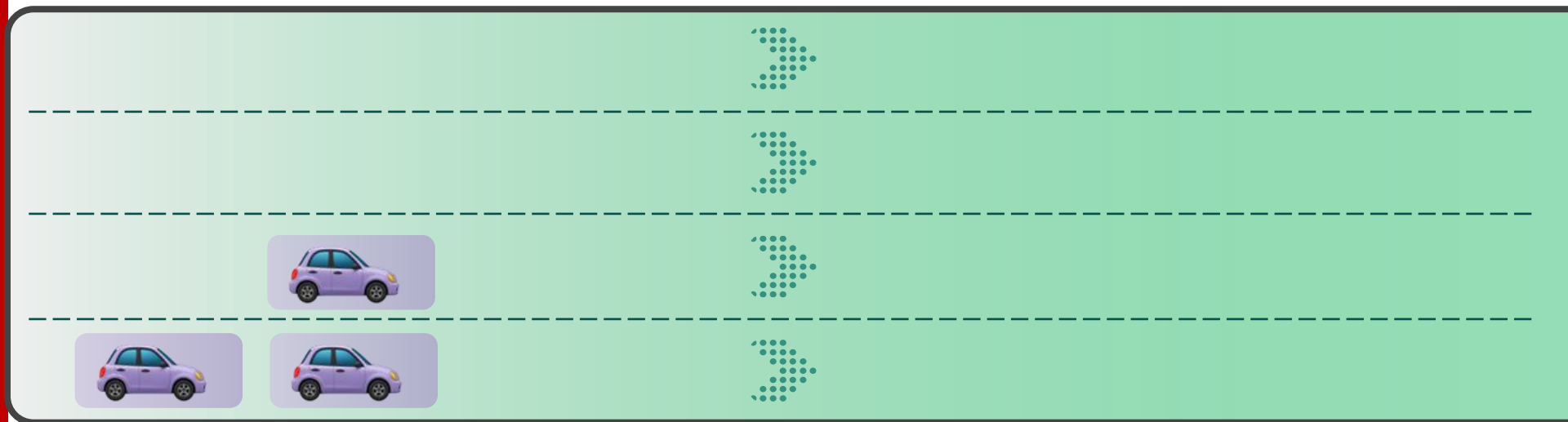
## 3. Scale

- Adds extra capacity while jobs keep running
- Immediate breathing room
- Capacity at least needs to be doubled



## 4. Switch capacities

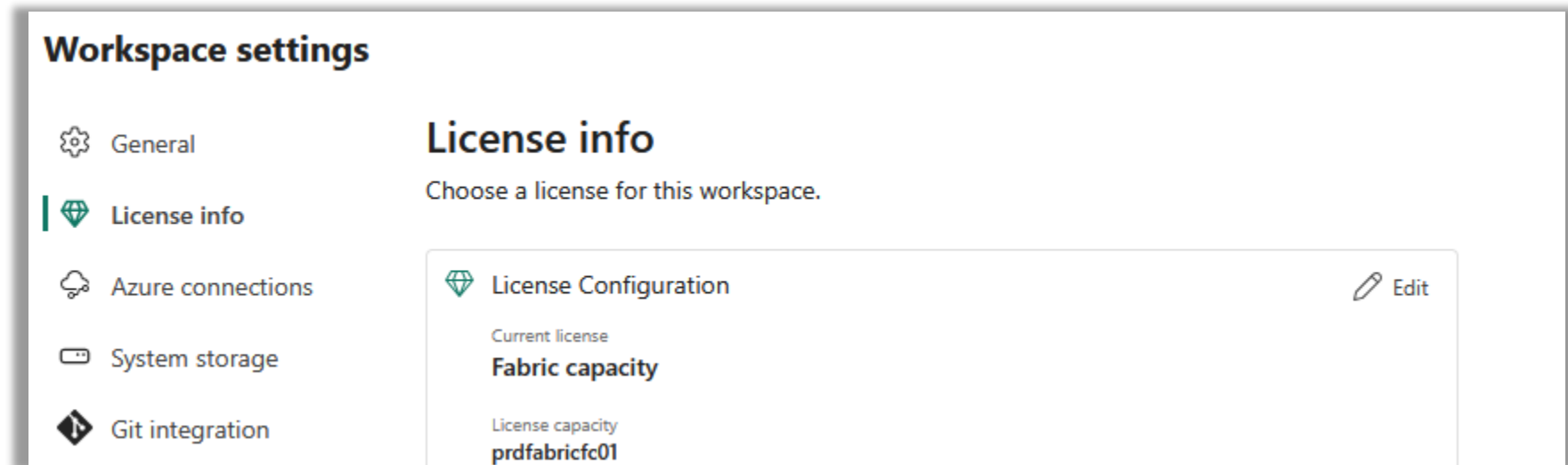
(Temporarily) move operations to a less busy highway to spread the usage



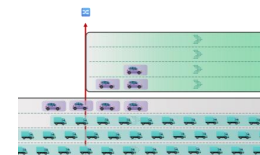
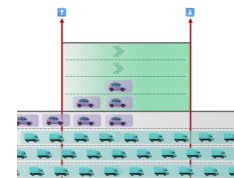
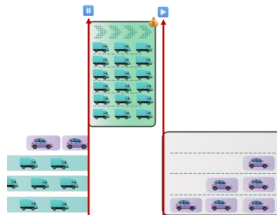
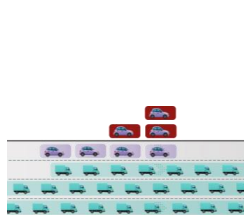
# Recovering a throttled capacity

## 4. Switch capacities

- Move workspace(s) to another, less busy capacity to reduce usage
- Capacity needs to be in the same region



|               | Wait ⌚                                                               | Pause & resume ⏸                                                               | Scale ⬆                                                          | Switch capacities 🔀                                         |
|---------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|
| <b>What</b>   | Sweat it out until capacity becomes available again                  | Pay-as-you go for overage                                                      | Scale the existing capacity                                      | Changing to another capacity                                |
| <b>Cost</b>   | None (patience)                                                      | Pay-as-you go for overage                                                      | At least double of the capacity                                  | Depending on capacity                                       |
| <b>Effect</b> | Can take up to 24hrs or longer                                       | Immediate one-time                                                             | Immediate temporary or permanent                                 | Immediate temporary or permanent                            |
| <b>Pro</b>    | No action needed<br>No cost                                          | Immediate effect<br>Start with clear capacity                                  | Running activities keep running / Gives plenty of breathing room | More flexible in choosing a capacity                        |
| <b>Con</b>    | Activities can not continue for a longer period of time = big impact | Running activities get cancelled / Pay for all smoothed usage can be expensive | Can be expensive                                                 | Only within same region & if other capacities are available |



# Story from the field



# Prevention



# Typical expensive tasks where you can start looking

## How to solve?

- Move workload to “cheaper” options
  - Dataflow gen2 → notebooks
  - Semantic model operations → transform in Warehouse or Lakehouse
  - Paginated reports → Power BI dashboards
- DAX optimisation:
  - Best Practice analyzer (in tabular editor or via semantic link labs)
  - If possible, move calculations towards Warehouse or lakehouse tables

# Prevention > Notifications

- Set notifications on your capacity settings when you hit certain thresholds

Notifications

Get notified when you're close to exceeding your available capacity (which includes base and Autoscale capacity)

Send notifications when

☒

You're using  % of your available capacity

☒

You've exceeded your available capacity and might experience slowdowns

☐

An Autoscale v-core has been added

☐

You've reached your Autoscale maximum

Send notifications to

☒

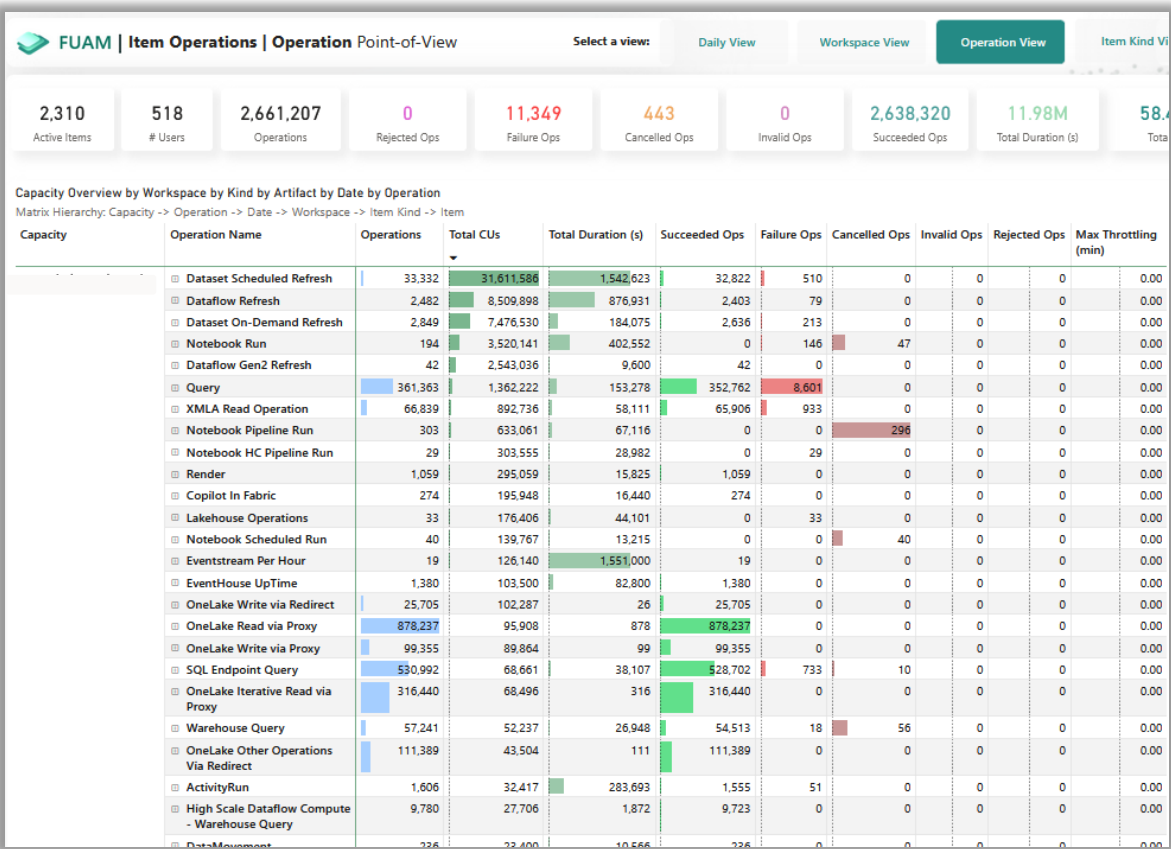
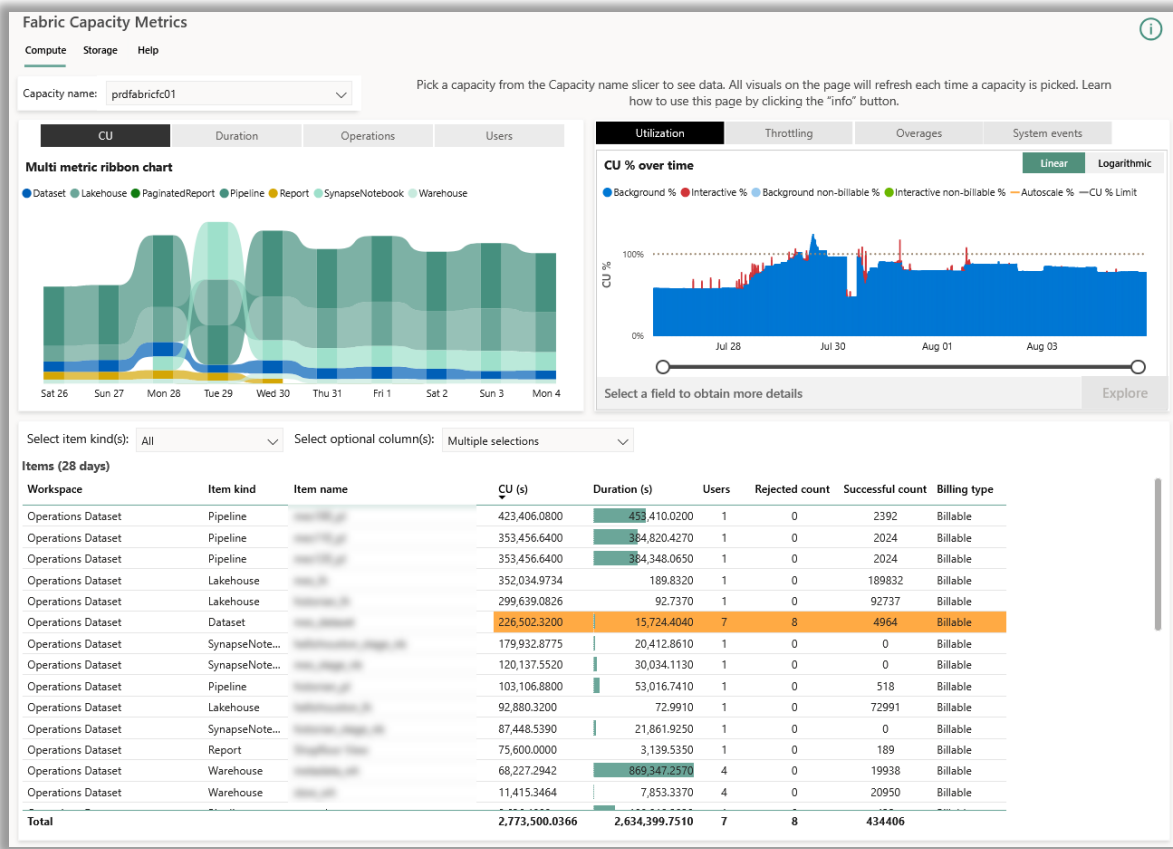
Capacity admins

☐

These contacts:

# Monitor & analyze bottlenecks

- Actively monitor using Capacity metrics app or FUAM



# Prevention > Capacity settings

## Surge protection

- Rejects new background operations to keep capacity for interactive operations
- Rejection Threshold
- Recovery Threshold
- Setting on capacity level (capacity admin)

Surge Protection

Unapplied changes

Background Operations

When total CU consumption reaches the rejection threshold, reject new background operation requests. When total CU consumption drops below the recovery threshold, accept new background operation requests.

☒ On

Rejection threshold

%

Recovery threshold

%

Apply

Discard

# Prevention > Capacity settings

## Surge protection

Only interactive operations allowed on specific lanes



Except



# Prevention > Capacity settings

## (Autoscale) Billing for Spark Jobs

- Spark jobs use pay-as-you-go consumption outside your capacity
- Ideal to keep lower capacities performant
- Setting on capacity level (capacity admin)

### Autoscale Billing for Apache Spark

*Unapplied changes*

Turn on this setting to use a pay-as-you-go model for Apache Spark jobs. With autoscale turned on, you can define a maximum CU limit on Spark consumption. Bursting and smoothing aren't applied when autoscale is in effect.

Changes to this setting, including turning it on or off and reducing the maximum CU limit for Spark consumption, cancels all currently running jobs. [Learn more](#) 

☒ On

Maximum Spark CU consumption (1 CU = 2 Spark v-cores)



24  Capacity Units

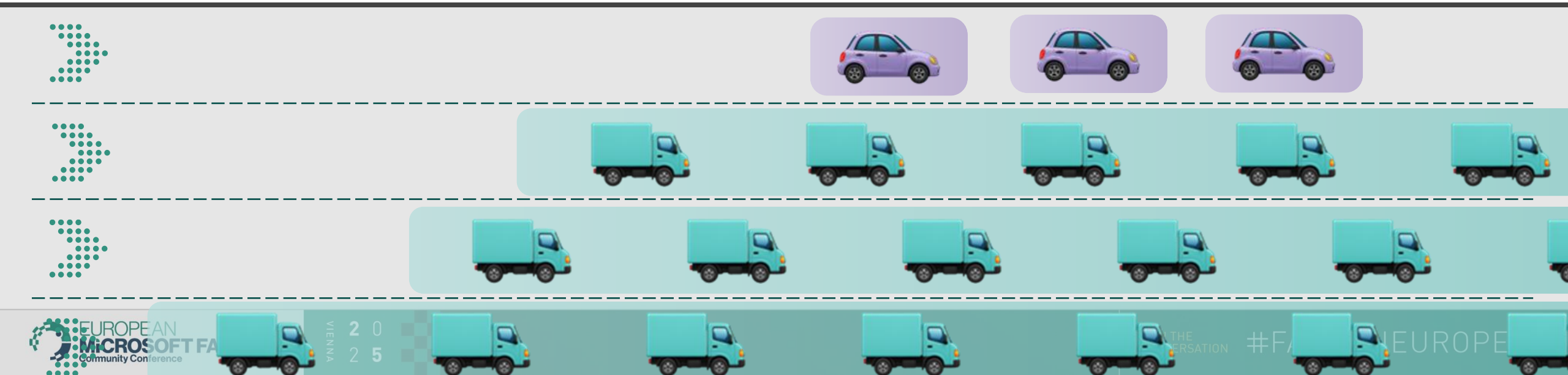
This setting uses 25% of your subscription's CU quota (32 of 128).  
Need more? [Learn how to increase your quota.](#)

# Prevention > Capacity settings



## (Autoscale) Billing for Spark Jobs

- Spark jobs use pay-as-you-go consumption outside your capacity



# Prevention > Capacity settings

## Copilot capacity

- Enable Fabric Copilot capacities in tenant settings
- Assign the capacity to specific users or groups



### Copilot and Azure OpenAI Service

▷ Users can use **Copilot** and other features powered by Azure OpenAI  
*Enabled for the entire organization*

▷ Data sent to Azure OpenAI can be processed outside your capacity's geographic region, compliance boundary, or national cloud instance  
*Enabled for the entire organization*

◀ Capacities can be designated as Fabric **Copilot** capacities  
*Enabled for the entire organization*

With this setting on, capacity admins can designate capacities as Fabric **Copilot** capacities. **Copilot** capacities are special capacity types that allow your organization to consolidate users' **Copilot** usage and billing on a single capacity. [Learn More](#)

When users use **Copilot** features, capacity admins can see the names of the items associated with users' **Copilot** activity. [Learn More](#)

☒ Enabled

Apply to:

☒ The entire organization

☐ Specific security groups

☐ Except specific security groups

Apply

Cancel

### ◀ Copilot capacity

*Enabled for a subset of the organization*

Turn on this setting to designate this capacity as a Fabric Copilot capacity. Copilot capacities are special capacity types that allow your organization to consolidate users' Copilot usage and billing on a single capacity. Copilot capacities may not be available in all regions. [Learn more](#)

Select the users or user groups who can use this capacity for their Copilot usage and billing.

Apply to:

☐ The entire organization

☒ Specific users or groups

Clear all

AT Armando Test ✕ Fabrice Admin ✕

Apply

Discard

# Prevention > Capacity settings



## Copilot capacity

Reroute all copilot requests to a specific capacity



# Prevention > Capacity settings



Surge protection

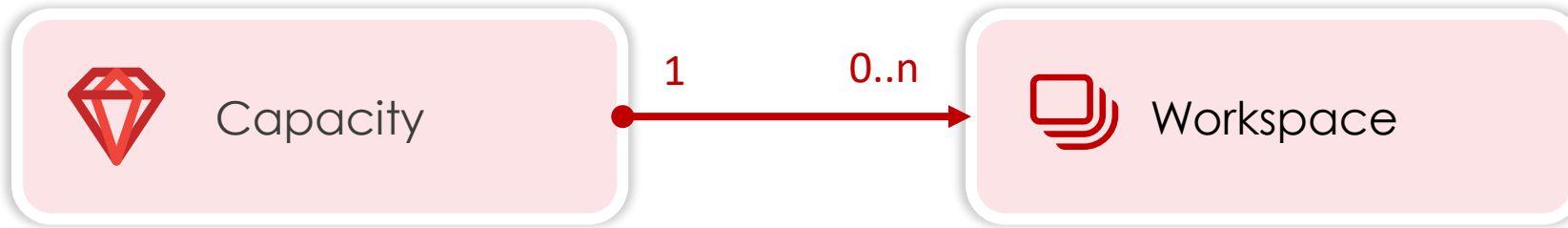


(Autoscale) Billing for Spark Jobs



CoPilot capacities

# Capacity planning



- F SKU (Fabric)
- Power BI Pro
- Power BI PPU
- Embedded

→ We are not limited to only Fabric SKU's. Power BI capacity is still a thing

# Capacity planning

*Rule of thumb*

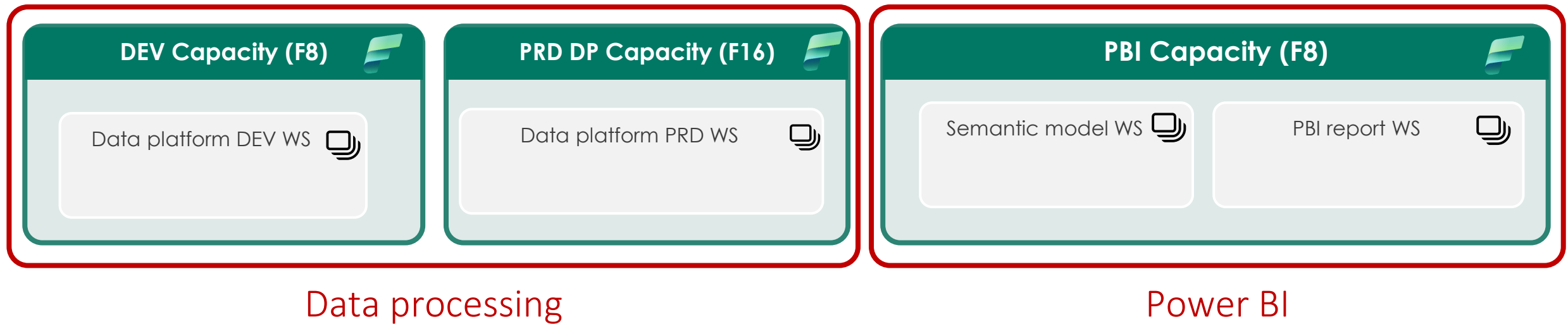
Split capacities between DEV & PRD



# Capacity planning

*Rule of thumb*

Split your data processing workload from your Power BI workload

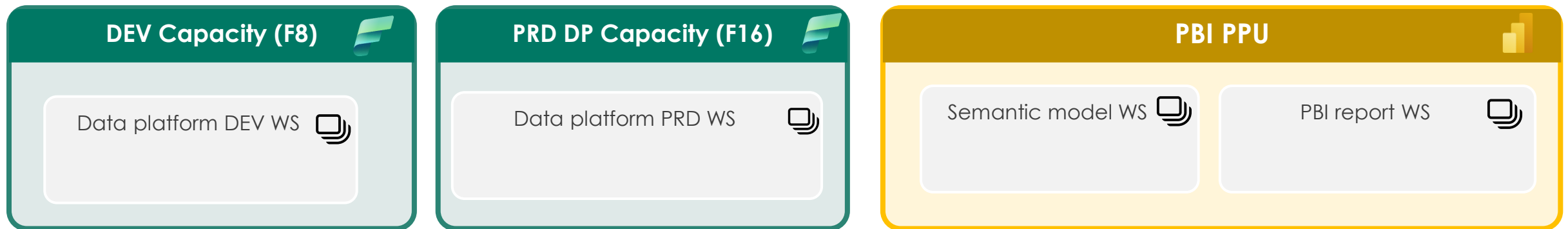


(keep in mind data model size limitations on smaller capacities - [link](#))

# Capacity planning

*Rule of thumb*

Know your options for Power BI

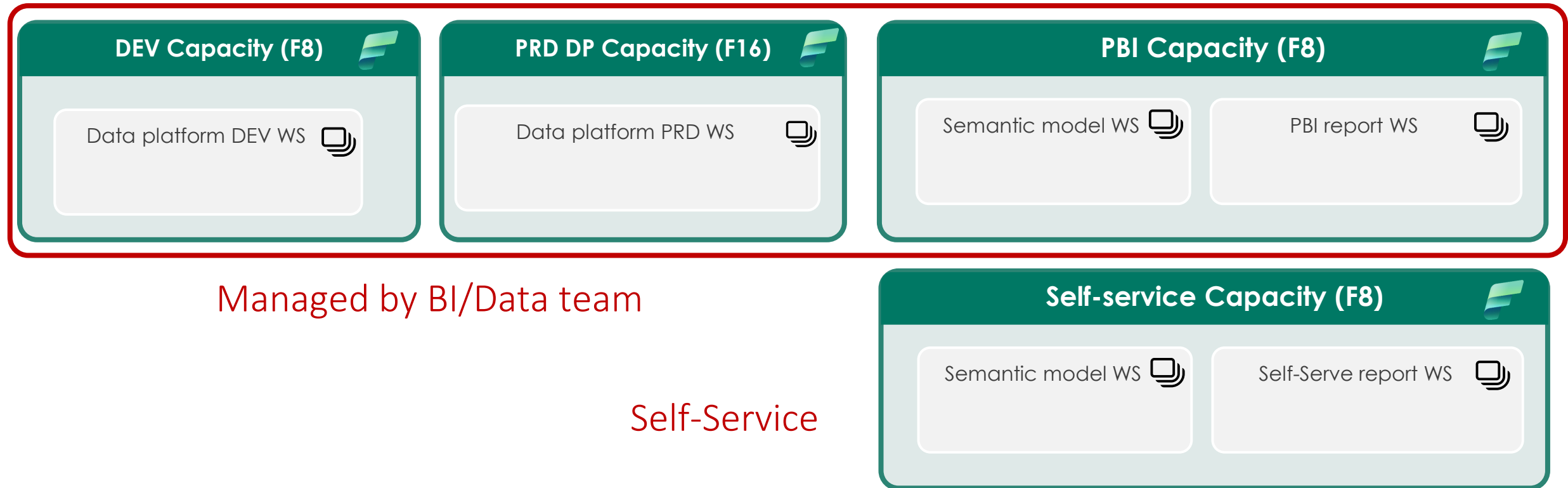


Premium Per User & Pro are still available and can be cost efficient

# Capacity planning

*Rule of thumb*

Split self-service from BI/Data team workloads



Managed by BI/Data team

Self-Service

# Capacity planning

*Rule of thumb*

Make sure your monitoring reports are not on your critical capacities

# Capacity planning

*Rules of thumb* 👉

1. Split capacities between DEV & PRD
2. Split your ETL workload from your PBI workload
3. Know your options for Power BI
4. Split self-service from BI/Data team workloads
5. Make sure your monitoring reports are not on your critical capacities

[Link to MS Capacity planning strategic guide](#)

# Wrap up

# We're at capacity, how could it come this far?

- smoothed background jobs  reduce capacity left for interactive operations 
- No more capacity to pay back for *overage* → throttling
- Multiple “grades” of throttling  
  - Interactive Delay
  - Interactive Rejection
  - Background Rejection

# We're at capacity, how do we identify the cause?



Check what is running & failing



Check the cause

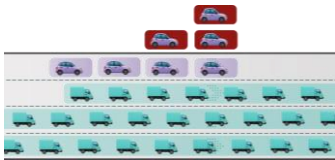


Detailed assessment of other  
risk factors to avoid future  
overruns

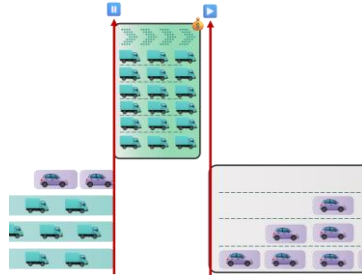


# We're at capacity, how do we recover operations?

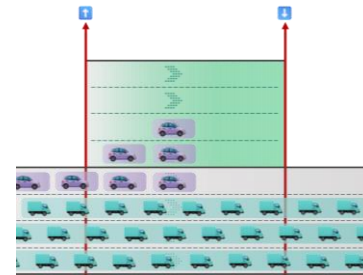
Wait ⌚



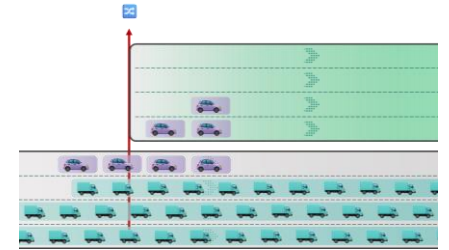
Pause & resume ▶||



Scale ↑



Switch capacities ↻



# We're at capacity, how do we prevent this in the future?

- Monitor & analyze for bottlenecks
- Get notified upfront
- Plan your workspaces & capacities in a smart way
  - Leverage multi-capacity flexibility
  - Remember Power BI PPU & Pro capacities
- Leverage special capacity options
  - Surge protection
  - Spark autoscale billing
  - Copilot capacities

# Capacity planning

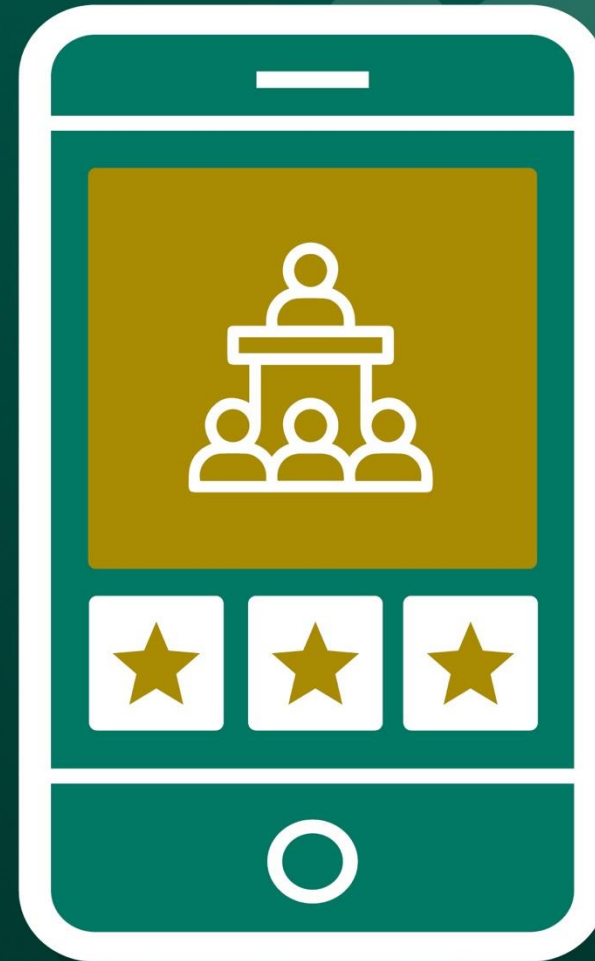
*Rules of thumb* 👉

1. Split capacities between DEV & PRD
2. Split your ETL workload from your PBI workload
3. Know your options for Power BI
4. Split self-service from BI/Data team workloads
5. Make sure your monitoring reports are not on your critical capacities

[Link to MS Capacity planning strategic guide](#)



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